

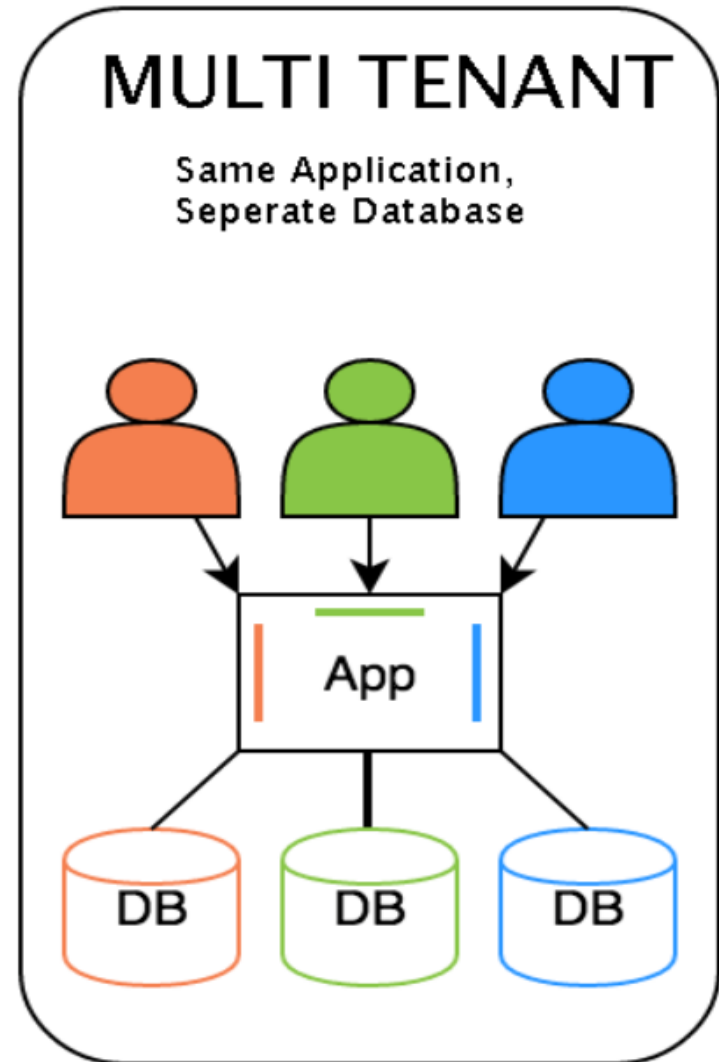
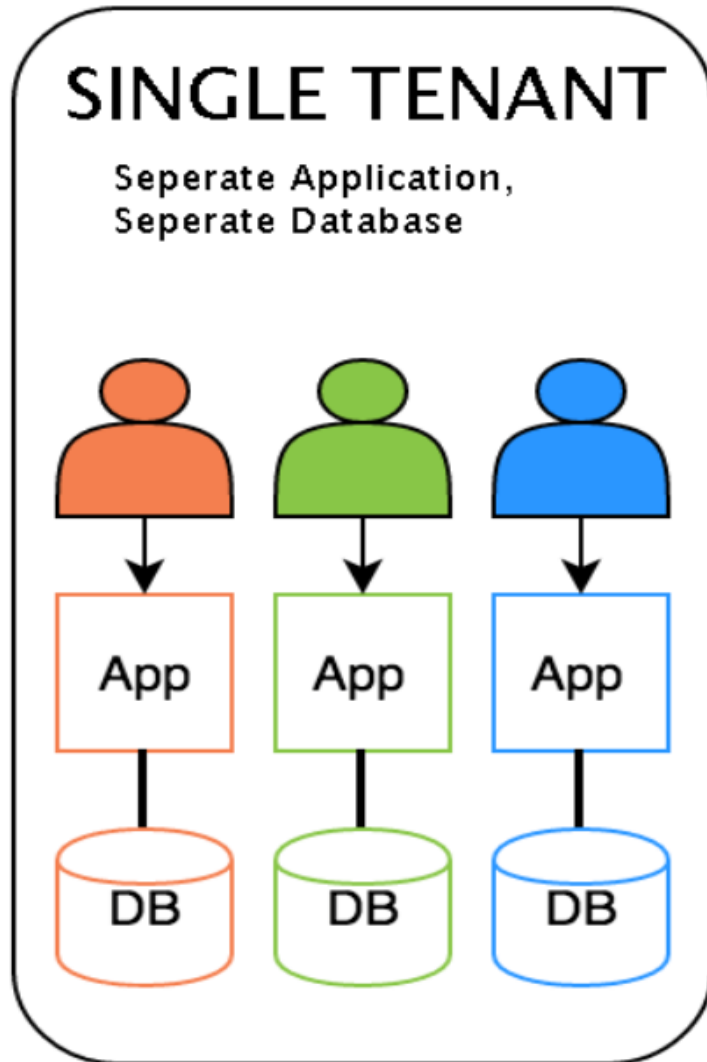
Cloud Computing

UNIT-II

Cloud Services and Platforms

- Cloud provides different types of services like
- Compute
- Storage
- Database
- Application
- Content delivery
- Analytics
- Deployment, management and identity and access management.
- *Infrastructure as a Service provides* virtualized dynamically scalable resources using a virtualized infrastructure.
- *PaaS* simplifies application development by providing development tools and APIs.
- *SaaS* provides multi-tenant applications hosted in the cloud.

SaaS provides multi-tenant applications hosted in the cloud.



Cloud Services and Platforms

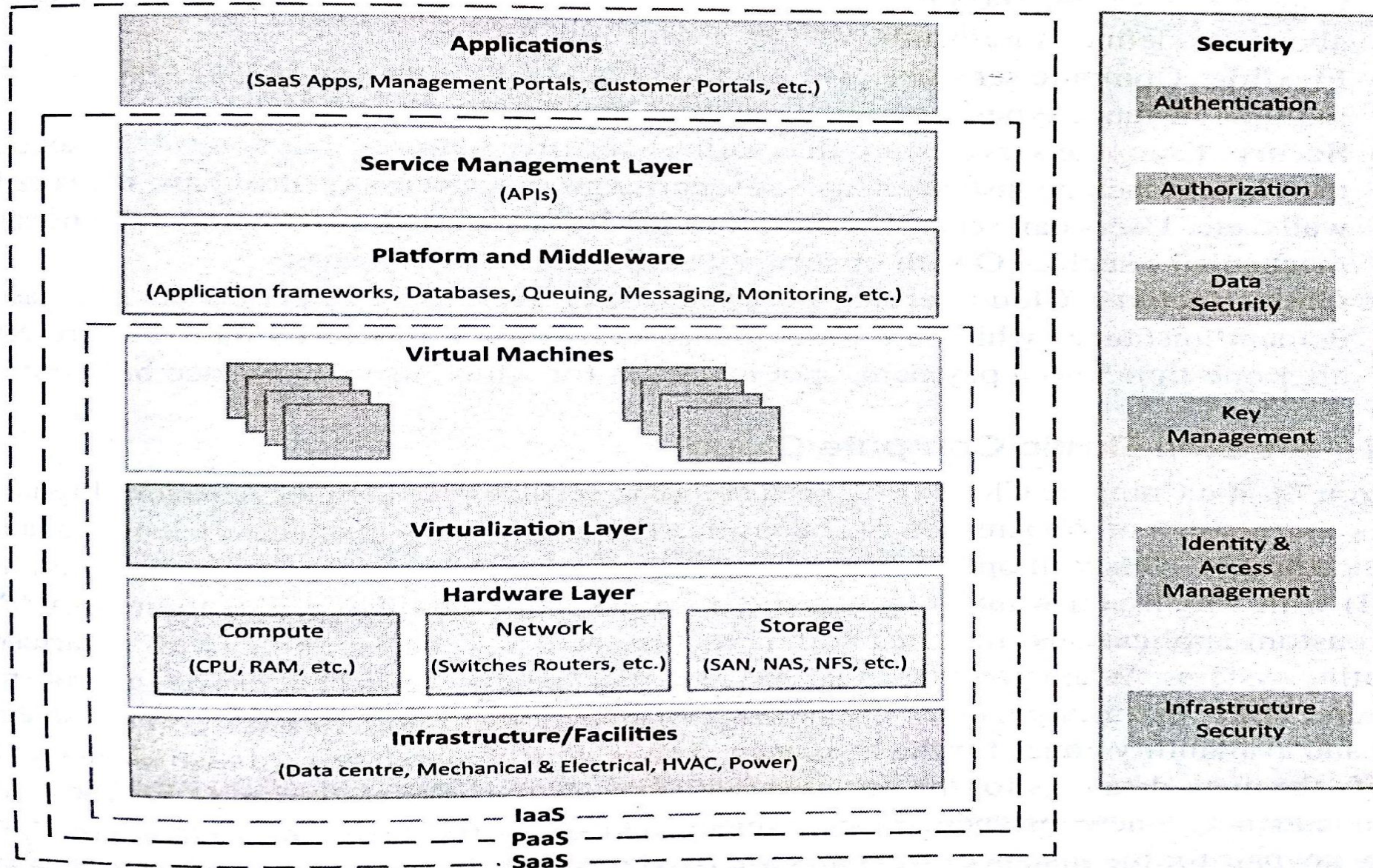
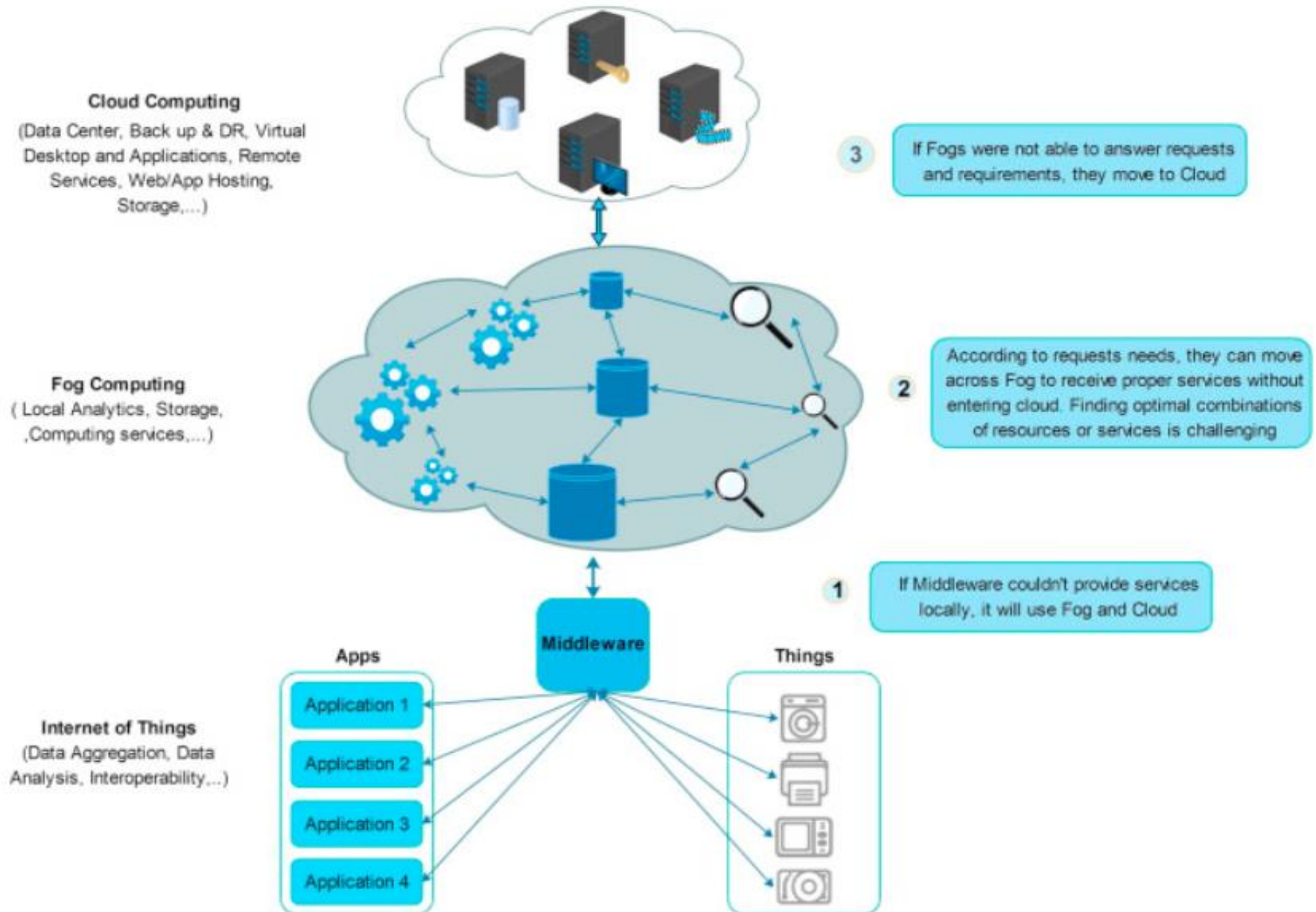


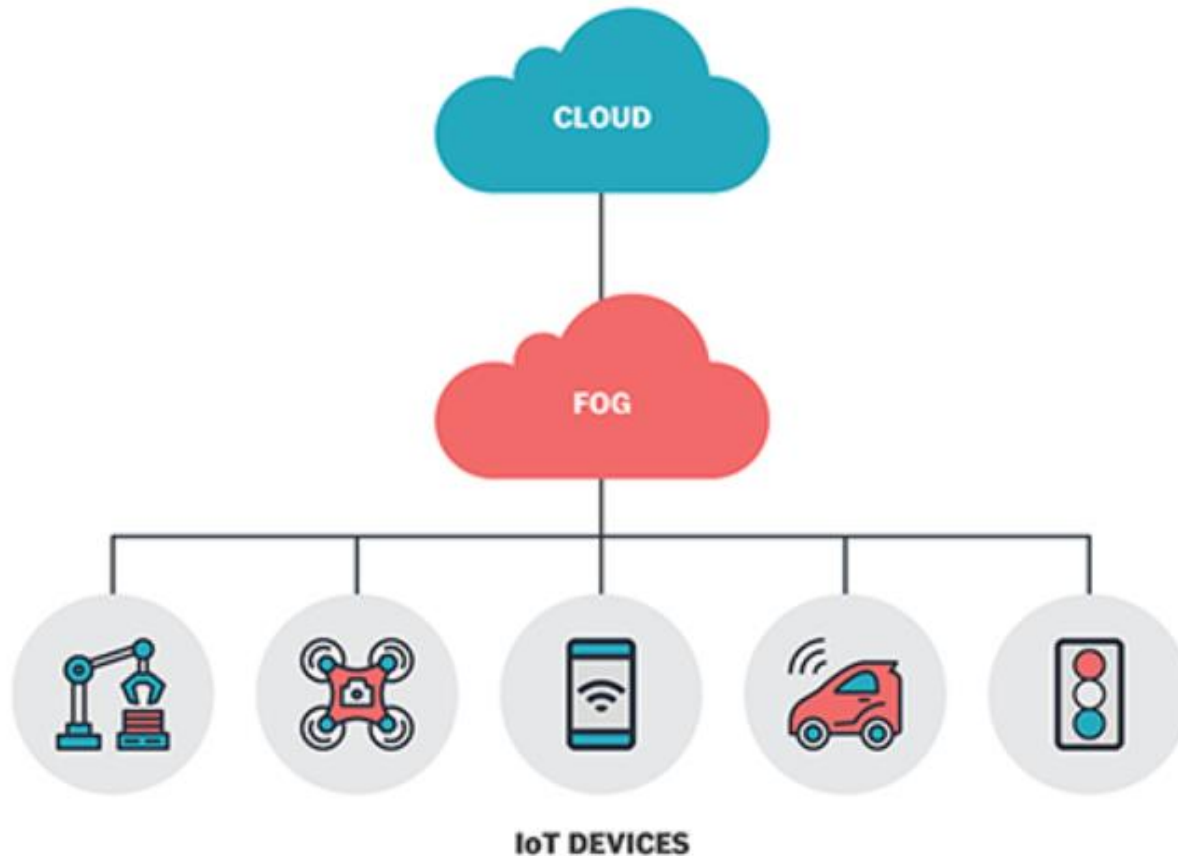
Fig : Cloud Reference model

Middleware



Fog computing is a decentralized computing infrastructure in which data, compute, storage and applications are located somewhere **between the data source and the cloud.**

Fog computing



Compute Services

- Compute services provide *dynamically scalable*(able to be changed in size) *compute capacity in the cloud* .
- Compute resources can be provisioned on-demand in the form of virtual machines.

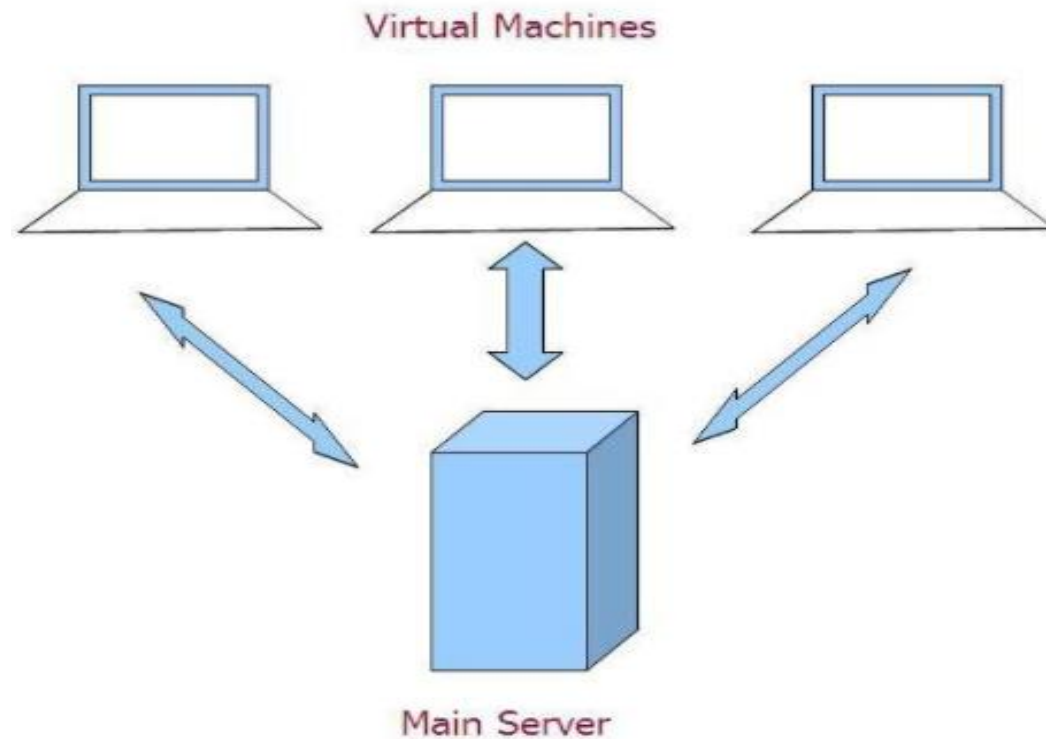


Fig: Virtual Environment

Virtual machine Advantages:

- Virtualization helps to achieve the isolation of applications. Suppose you have two Virtual Machines on your system even if one of them crashes *it will not effect the other virtual Machine.*
- You can have multiple operating system running on your machine simultaneously.
- It provides high availability of your applications. You can cluster your applications, even if the application on one Virtual Machine goes down, application on the other Virtual Machine will keep on running seamlessly(smoothly and continuously,).

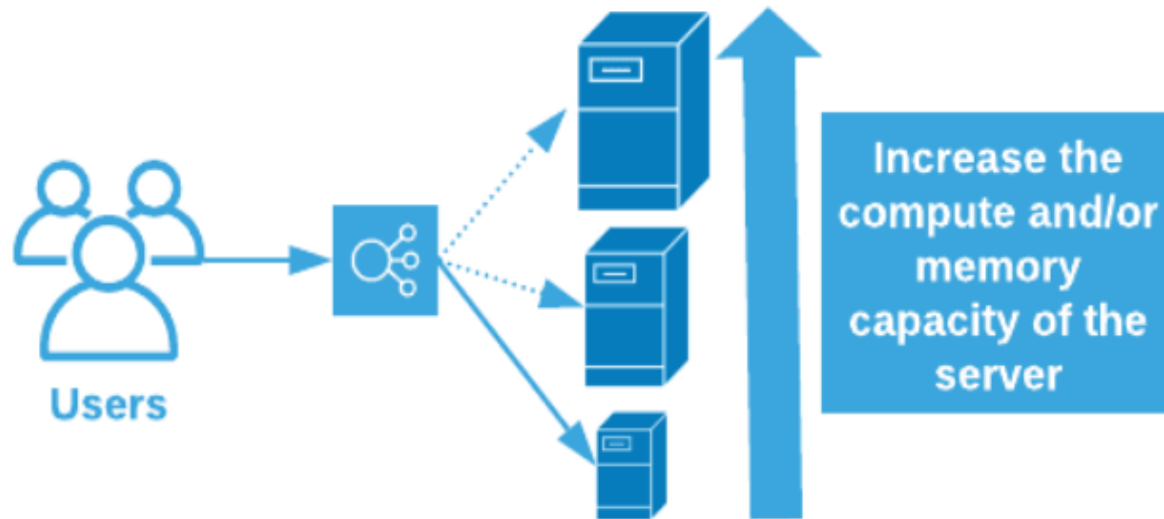
Compute Services

Features :

- **Scalable:** Compute services allow rapidly provisioning as many virtual machine instances as required.

1. Vertical scale or scale-up:

Adding more **compute or memory resources** for your applications increases the maximum capacity of the server. When demand increase, there should not be any noticeable change to your applications.



2. Horizontal scale or scale-out:

Adding more **Individual servers** into a resource pool where your applications run.



Compute Services

Features :

- **Flexible:** compute services give a wide range of options for virtual machines with multiple instance types, operating systems, zones/regions etc.



- It has made **employees more flexible**, they may complete their tasks in and out of the workplace. **Staff may easily access all important files through web browser.**
- There is no restriction on **devices and location**. It simultaneously shares business documentation and different other files.

Compute Services

Features :

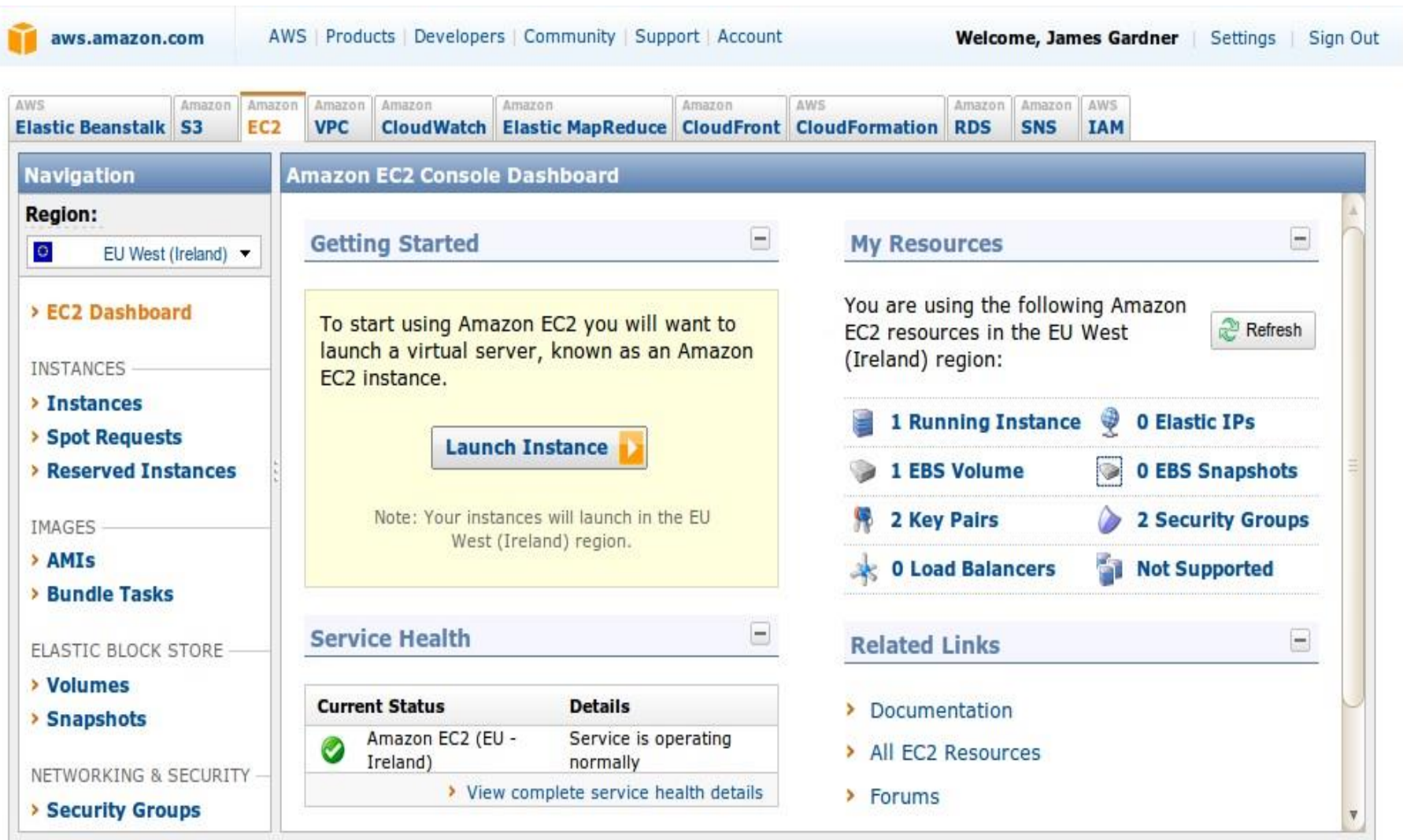
- **Secure:** It provides various security features that control the access to the virtual machine instances such as **security groups, access control lists, network firewalls etc.**
- **Cost effective:** *Cloud service providers* (CSP) offer various billing options such as on-demand, reserved , spot instances.

Compute Services

1. Amazon Elastic Compute Cloud :

- Amazon EC2 is a compute **service provided by amazon.**
- By Click on the launch instance button we **create a new instances**(***single occurrence of something.***) **with the user choice.**
- When you launch the instance you can specify the instance type with the number of instances to launch based on the selected **Amazon Machine Image** AMI and availability zones for the instance.
- At the time launch instance the user selects a key-pair from existing key pairs or create new **one to provide secure connection after it launch.**
- Security groups are used to open or block a specific network port for the launched instances.
- By using public DNS connect to the instances using SSH.

Cloud Services Examples



The screenshot displays the Amazon EC2 Console Dashboard. At the top, the AWS logo and navigation links are visible. The main header shows the user's name, 'James Gardner', and options for 'Settings' and 'Sign Out'. Below the header, a row of service links includes Elastic Beanstalk, S3, EC2 (highlighted), VPC, CloudWatch, Elastic MapReduce, CloudFront, CloudFormation, RDS, SNS, and IAM. The left sidebar contains a 'Navigation' menu with sections for 'Region' (set to EU West (Ireland)), 'EC2 Dashboard', 'INSTANCES' (with links to Instances, Spot Requests, and Reserved Instances), 'IMAGES' (with links to AMIs and Bundle Tasks), 'ELASTIC BLOCK STORE' (with links to Volumes and Snapshots), and 'NETWORKING & SECURITY' (with a link to Security Groups). The main content area is titled 'Amazon EC2 Console Dashboard' and is divided into four panels. The 'Getting Started' panel provides instructions on launching a virtual server and includes a 'Launch Instance' button. The 'My Resources' panel shows a summary of resources in the EU West (Ireland) region, including 1 Running Instance, 0 Elastic IPs, 1 EBS Volume, 0 EBS Snapshots, 2 Key Pairs, 2 Security Groups, 0 Load Balancers, and Not Supported. The 'Service Health' panel displays the current status of Amazon EC2 (EU - Ireland) as 'Service is operating normally'. The 'Related Links' panel provides links to Documentation, All EC2 Resources, and Forums.

Navigation

Region: EU West (Ireland)

EC2 Dashboard

INSTANCES

- Instances
- Spot Requests
- Reserved Instances

IMAGES

- AMIs
- Bundle Tasks

ELASTIC BLOCK STORE

- Volumes
- Snapshots

NETWORKING & SECURITY

- Security Groups

Amazon EC2 Console Dashboard

Getting Started

To start using Amazon EC2 you will want to launch a virtual server, known as an Amazon EC2 instance.

Launch Instance

Note: Your instances will launch in the EU West (Ireland) region.

My Resources

You are using the following Amazon EC2 resources in the EU West (Ireland) region:

- 1 Running Instance
- 0 Elastic IPs
- 1 EBS Volume
- 0 EBS Snapshots
- 2 Key Pairs
- 2 Security Groups
- 0 Load Balancers
- Not Supported

Service Health

Current Status	Details
 Amazon EC2 (EU - Ireland)	Service is operating normally

[View complete service health details](#)

Related Links

- Documentation
- All EC2 Resources
- Forums

Fig : Amazon EC2 Dashboard

Compute Services

2. Google Compute Engine

- It is a compute **service provided by google**.
- GCE allows the user to create and manage compute instances.
- To create a new instance the user selects an instance machine type, zone, **machine image and name of instance, instance tags and meta-data**.
- Depending on instance type it creates two disk resources like scratch disk space and persistent disk space.
- By default, traffic between instances in the same network, over any port and any **protocol and incoming SSH connections from anywhere are enabled**.

Cloud Services Examples

Google Cloud Console

GCE Demo Project
gce-io-demo

Compute Engine

NEW INSTANCE

Instances

Disks

Snapshots

Images

Networks

Metadata

Zones

Operations

Quotas

Create a new Instance

Name ⓘ lowercase, no spaces

Description ⓘ
Optional

Tags ⓘ comma separated
Optional

Metadata key value - +

Location and Resources

Zone us-central1-a

Machine Type n1-standard-1

Boot Source ⓘ New persistent disk from image

Image debian-7-wheezy-v20130509

Additional Disks ⓘ
Optional No disks in zone us-central1-a

Summary

Enter a name

debian-7-wheezy-v20130509
Debian GNU/Linux 7.0 (wheezy) b...

us-central1-a

1 vCPU 3.75 GB RAM

default
Default network for the project

Create Discard

Equivalent [REST](#) or [command line](#)

Fig : Google Compute Engine Dashboard

Compute Services

3. Windows Azure Virtual Machines

- It is the compute **service from Microsoft**.
- To create a new instance, you select the instance type and the machine image.
- You can either provide a **user name and password** or upload a certificate file for securely connecting to the instance.
- Any changes made to the VM are persistently stored and new VMs can be created from the previously stored machine images.

Cloud Services Examples

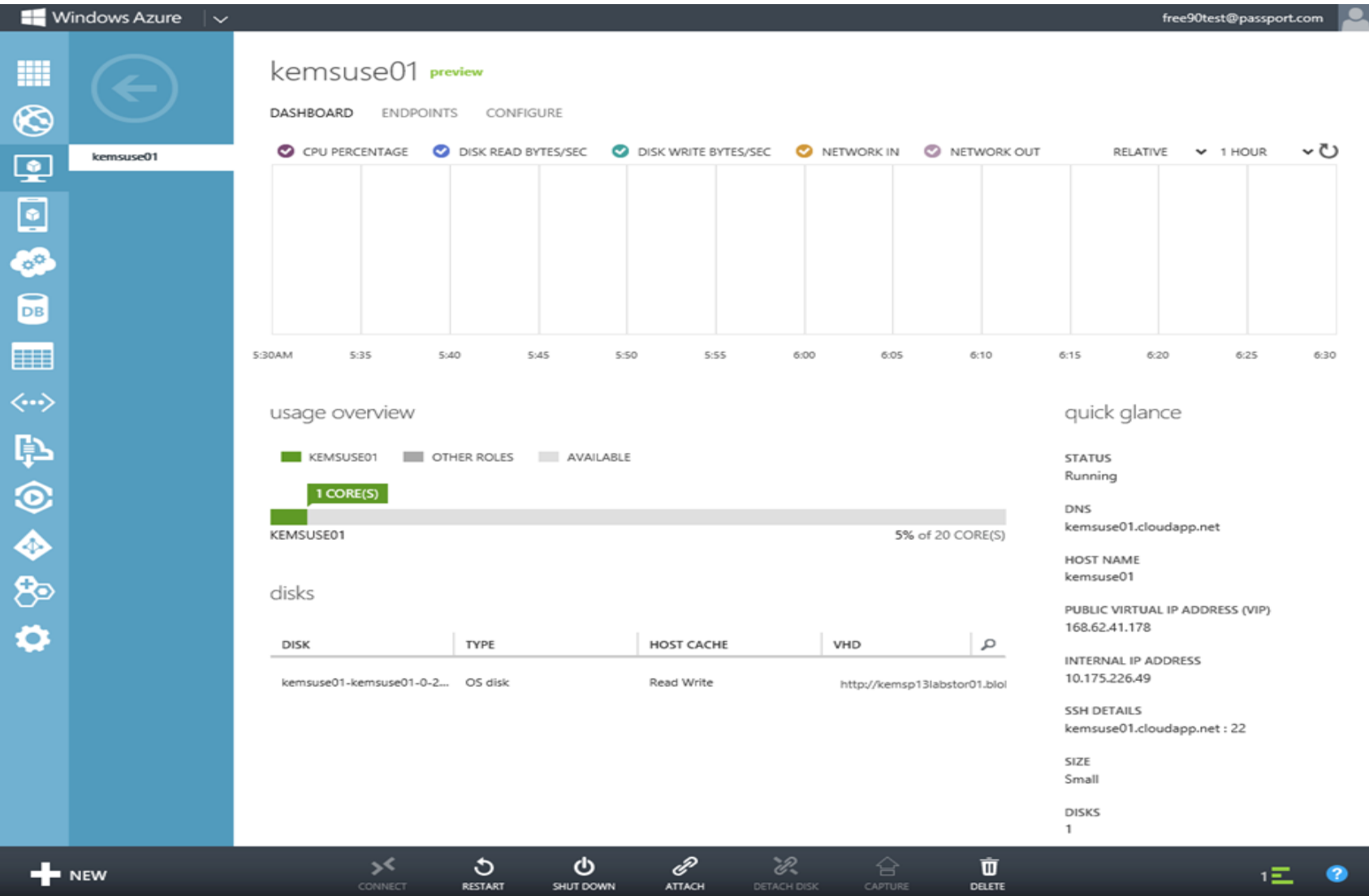


Fig : Windows Azure VM dashboard

Storage Services

- Cloud storage services allow storage and retrieval of any amount of data, at any where on the web.
- Data in cloud storage services is organized into ***buckets*** or containers.

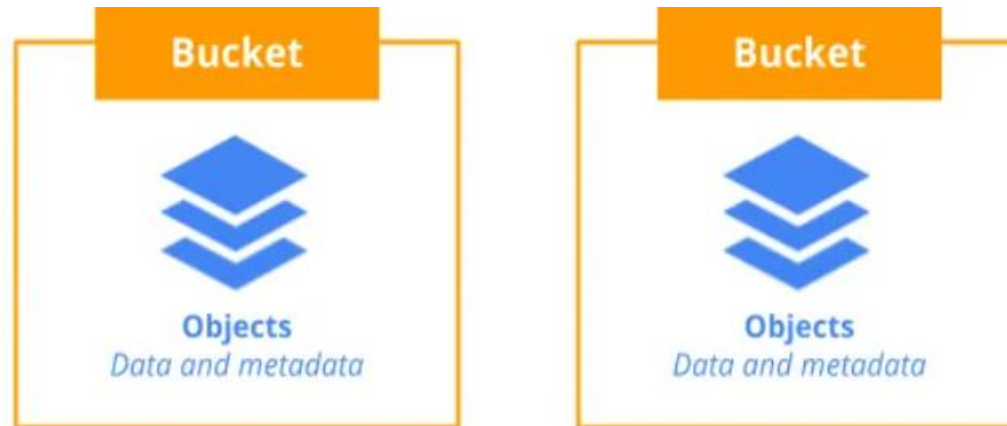
Features :

- **Scalability** : Cloud storage services provide high capacity and scalability(***capacity to be changed in size***).
- Objects upto several **tera-bytes** in size can be uploaded and multiple buckets or containers can be created on cloud storage.
- **Replication** : When an object is uploaded it is replicated at multiple facilities on multiple devices within each facility.
- **Access Policies** : It provide Access Control Lists and Bucket/container level policies to provide security.
- **Encryption** : It provide Server side encryption option to encrypt all data stored in cloud storage.
- **Consistency**: Strong data consistency is provided for all uploaded and delete operations.

Storage Services

Amazon Simple Storage Service : (S3)

- Amazon *simple storage service*(S3) is an online cloud based data storage infrastructure for **storing and retrieving any amount of data**.
- S3 provides **highly reliable, scalable, fast**, fully redundant and affordable storage infrastructure.
- Data stored on S3 is organized in the form of buckets.
- It is necessary to create a bucket before you can store data on S3.
- You can upload any **kind of file to S3**.
- While uploading a file, you can specify the redundancy and encryption options and access permissions.



Storage Services

 **AWS Management Console** › **Amazon S3** 12/23/2011 17:58 GMT+0000 | [Help](#)

AWS Elastic Beanstalk **Amazon S3** **Amazon EC2** **Amazon VPC** **Amazon CloudWatch** **Amazon Elastic MapReduce** **Amazon CloudFront** **AWS CloudFormation** **Amazon RDS** **Amazon ElastiCache** **Amazon SQS** **AWS IAM** **Amazon SNS** **Amazon SES** **Amazon Route 53**

Buckets

 Create Bucket **Actions** ▾

 **beaglebuntubackup**

Objects and Folders

 Upload  Create Folder **Actions** ▾  Refresh  Properties  Transfers  Help

 **beaglebuntubackup**

Name	Size	Last Modified
 beaglebuntu-20111223.md5	151 bytes	Fri Dec 23 17:58:23 GMT+000 2011
 beaglebuntu-20111224.md5	137 bytes	Sat Dec 24 06:25:42 GMT+000 2011
 beaglebuntu-etc.20111223.master.tar.gz	650.9 KB	Fri Dec 23 17:58:23 GMT+000 2011
 beaglebuntu-etc.20111224.tar.gz	15.8 KB	Sat Dec 24 06:25:42 GMT+000 2011
 beaglebuntu-home-svn.20111223.master.tar.gz	24.2 MB	Fri Dec 23 17:58:38 GMT+000 2011
 beaglebuntu-home-svn.20111224.tar.gz	1.5 KB	Sat Dec 24 06:25:43 GMT+000 2011

Fig : Amazon S3 Console

Storage Services

Google Cloud Storage

- Objects in GCS are organized into buckets.
- ACLs are used to control access to objects and buckets.
- ACLs can be configured to share objects and buckets with the entire world, a google group, a google-hosted domain or specific google account holders.

The screenshot shows the Google Cloud Console interface. At the top, the header reads "Google Cloud Console" with a user email "@gmail.com" and links for "Customer support" and "Sign out". Below the header, the left sidebar shows "Cloud" with "Project ID: cloud" and "Cloud Storage" selected. The main area displays the "cloudbucket" interface. It includes buttons for "Upload" and "New Folder", a "Filter by prefix..." search bar, and a table of objects. The table has columns for NAME, SIZE, TYPE, LAST UPLOADED, and SHARED PUBLICLY. Five objects are listed: Application.xml (7.08KB, text/xml), Screenshot.png (222.96KB, image/png), cost.xls (16KB, application/vnd.ms-excel), dash.html (13.62KB, text/html), and index.html (7.06KB, text/html). Each row has a checkbox for sharing publicly.

	NAME	SIZE	TYPE	LAST UPLOADED	SHARED PUBLICLY
☐	Application.xml	7.08KB	text/xml	Aug 16, 2013 2:47:40 PM	☐
☐	Screenshot.png	222.96KB	image/png	Aug 16, 2013 2:47:53 PM	☐
☐	cost.xls	16KB	application/vnd.ms-excel	Aug 16, 2013 2:47:42 PM	☐
☐	dash.html	13.62KB	text/html	Aug 16, 2013 2:47:44 PM	☐
☐	index.html	7.06KB	text/html	Aug 16, 2013 2:47:46 PM	☐

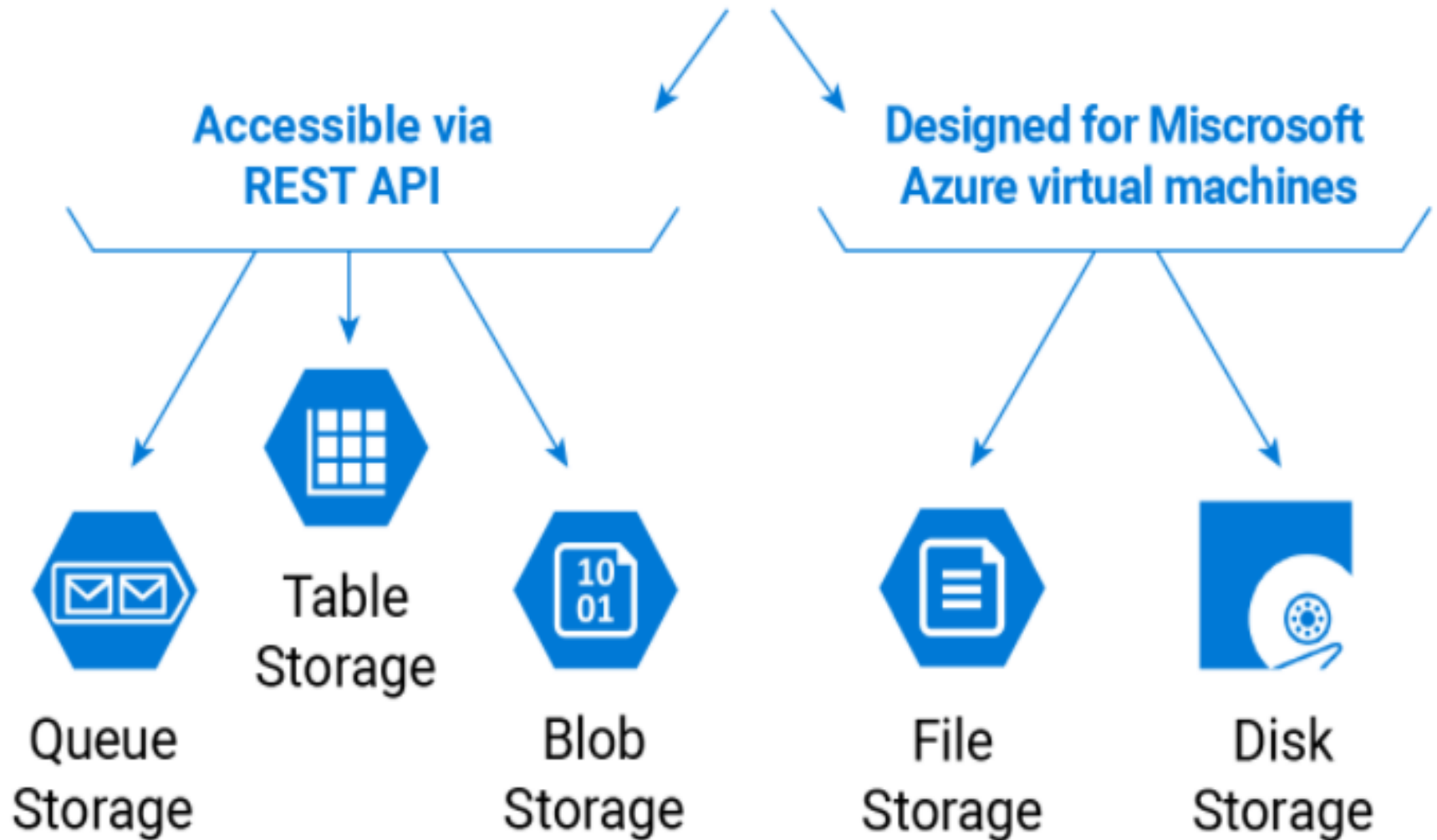
Storage Services

Windows Azure Storage

- It is a cloud storage **service from Microsoft**.
- Windows Azure Storage provides various storage service such as *blob storage service, table service and queue service*.
- The blob storage service allows strong unstructured binary data or binary large objects.
- Blobs are organized into containers.
- There are two kinds of blobs can be stored – **block and page blobs**.
- A block blob can be subdivided into some number of blocks.
- **If a failure occurs while transferring a block blob**, retransmission can resume with the most recent block rather than sending the entire blob again.
- Page blobs are divided into number of pages and are designed for random access.
- Applications can read and write individual pages at random in a page blob.

Storage Services

Microsoft Azure Storage



Storage Services

Database Services

Windows Azure

CREDIT STATUS

portalvhdsdw1v14g2x7kb7

DASHBOARD MONITOR CONFIGURE CONTAINERS

RELATIVE 6 HOURS

Monitoring chart unavailable. Enable monitoring in the configure page.

9:30AM 10:30 11:30 12:30PM 1:30 2:30 3:30

services

SERVICE	ENDPOINT
Blobs	http://portalvhdsdw1v14g2x7kb7.blob.core.windows.net/
Tables	http://portalvhdsdw1v14g2x7kb7.table.core.windows.net/
Queues	http://portalvhdsdw1v14g2x7kb7.queue.core.windows.net/

quick glance

STATUS
Online

GEO-REPLICATION
Enabled

LAST GEO-REPLICATION FAILOVER
Never

PRIMARY REGION
West US

SECONDARY REGION
East US

CUSTOM DOMAIN
None

LOCATION
West US

CREATED
8/16/2013 3:17:21 PM

Fig : Windows azure Storage Console

Database Services

- Cloud database services allow you to set-up and operate relational or non-relational data bases in the cloud.
- The benefit of using cloud database services is that it relieves the application developers from the time consuming database administration tasks.
- Relational databases provides by cloud services are like *MySQL, Oracle, SQL Serve etc.*
- The non relational data bases provided by cloud service providers are mostly proprietary solutions, No-SQL databases are usually fully-managed and deliver seamless throughput and scalability.

Features :

- Provisioned capacity can be scaled-up or down.
- For Heavy work loads, read-replicas can be created.

Database Services

- **Reliability** : Cloud database services are reliable and provide automated backup and snapshot options.
- **Performance** : Cloud database services provide guaranteed performance with options such as guaranteed *input/output operations per second (IOPS)* which can be provisioned upfront.
- **Security** : Cloud database services provide several security features to restrict the access to the database instances and stored data, such as network firewalls and authentication mechanisms.

Amazon Relational Data Store:

- Amazon **relational database service(RDS)** is a web service that makes it easy to setup, operate and scale a relational database in cloud.
- The console provides an instance launch wizard that allows you to select the type of database to **create, database instance size, allocated storage, DB instance identifier, DB username and Password.**
- After creating the instance it create an end point, using this end point you can connect instance securely.

Database Services

Amazon DynamoDB :

- Amazon Dynamo DB is the **non-relational database service** from Amazon.
- **This model includes table, attributes and items.**
- A table is a collection of items and each item is a collection of attributes.
- To store data you need to ***create one or more tables*** and specify the required throughput capacity you want to provision and reserve for read and writes.
- It is a fully managed service that automatically spreads the stored tables to number of servers to meet the throughput requirements specified by the user.
- All the stored data is automatically replicated across multiple availability zones to provide data durability.

Database Services

Google Cloud SQL:

- It is a relational database from Google.
- It allows you to host MYSQL databases in the goolges cloud.
- It provides the ability to import and export the databases.
- To create new instance you select a region, database tier, billing plan and replication mode.
- You can schedule daily backups for your Google Cloud SQL instances, and also restore back-up databases.

Google Cloud Datastore:

- It is a fully managed non-relational database from google.
- It offers ACID transactions and high availability of read and writes.
- It consists of entities and each entity has one or more properties(Key value pairs) which can be of one of several supported datatypes like string and integer.
- The entity kind is used for categorizing the entity for the purpose of queries and entity key uniquely identifies the entity.

Database Services

Windows Azure SQL Databases

- Windows azure SQL database is the relational database service from the Microsoft.
- **Azure SQL Database is based on the SQL server.**
- It does not give each customer a separate instance of SQL server. Instead the Database is a multi-tenant service, with a logical SQL database server for each customer.

Windows Azure Table Services

- It is a **non-relational(No-SQL) database service from Microsoft.**
- It consists of tables having multiple entities.
- Tables are divided into some number of partitions, each of which can be stored on a separate machine.
- Each partition in a table holds a specified number of entities, each containing as many as 255 properties.
- Each property can be one of the several supported data types such as integers and strings.

Application Services

- There are **different types of cloud application** services like application *runtime and frameworks*. *queuing services, email services, notification services and media services.*

1. Application Runtime and Frameworks:

- *It allows the developers to develop and host applications in the cloud.*
- **Application runtime** provide support for **programming languages** and automatically allocate resources for applications without the need to run and maintain the server.

Google App Engine(GAE)

- It is a ***PaaS*** from google and it includes both an application and web frameworks.

Features :

- **Runtimes:** App Engine supports applications developed in ***java, python and PHP programming languages*** and it provides runtime environment for those languages.

Application Services

- **Sandbox** : Applications run in a secure sandbox(**testing environment**) environment isolated from other applications.
- **Web Frameworks**: App engine provides a simple **python** web application framework called webapp2.
- **Data store**: App Engine provides a **no-SQL** data storage services.
- **Authentication**: App Engine applications can be integrated with google accounts for user authentication.
- **URL Fetch Service**: URL Fetch service allows applications to access resources on the **internet like web services**.
- **Email Services**: Email service allows applications to send email messages.
- **Image Manipulation service**: Image manipulation service allows applications to **resize, crop, rotate, flip and enhance Images**.
- **Memcache**: Memcache service is **high performance** in-memory key-value cache service that applications can use for caching data items that do not need a persistent storage.

Application Services

- **Task Queues:** Task queues allow applications to do work in the *background by breaking up work into small*, discrete units called tasks which are enqueued in *task queues*.
- **Scheduled Tasks Service:** App Engine provides a Cron service for scheduled tasks that *trigger* events at specified times and regular intervals.

Windows Azure Web Sites:

- It is a **PaaS** service from Microsoft.
- It allows you to *host web applications* in the azure cloud.
- Azure web sites provide ***shared and standard options***.
- In the shared option, It run on a set of virtual machines that may contain ***multiple websites created by multiple users***.
- In the standard option, ***Azure web sites run on virtual machines that belongs to an individual user***.
- It support web applications developed in ***ASP.NET,PHP,NODE.js and python languages***.

Application Services

2. Queuing Services :

- Cloud-based queuing services allow *de-coupling* application components.
- The de-coupled components communicate via messaging queues.
- Queues are useful for *asynchronous processing*.
- Queues act as overflow buffers to handle temporary volume spikes (thin) or mismatches in message generation and consumption rates from application components.

Amazon simple Queue Service

- It is a *queuing service* from Amazon.
- *Simple Queue Service* (SQS) is a distributed queue that supports messages of up to 256 KB.
- SQS supports multiple writers and readers and locks messages while they are being processed.
- To ensure high availability for delivering messages, SQS service trades off on the first in first out capacity and does not guarantee that messages

Application Services

- Applications that *require FIFO ordering* of messages can place additional sequencing information in each message so that they can be re-ordered after retrieving from a queue.

Google Task Queue Service

- It is a *queuing service from google* and it is part of the Google App Engine platform.
- Task queue allow applications to execute in *background*.
- Task is unit of work to be *performed by an application*.
- Task objects consists of application-specific URL with a request handler for the task, and an optional data payload that parameterizes the task.
- There two configurations for Task Queues *1. Push Queue 2.Pull Queue*.
- Push queue is default *queue that processes tasks based on the processing rate configured* in the queue definition.

Application Services

- **Pull queues** allow task consumers to lease a specific number of tasks for a specific definition.
- The tasks are processes and deleted before the lease ends.

Windows Azure Queue Service

- It is a **queuing service from Microsoft**.
- Azure queue **service allows storing large numbers of messages** that can be accessed from anywhere in the world via authenticate calls using HTTP and HTTPS.
- The size of single message can be up to 64KB.

Application Services

3. Email Services

- Cloud based email services allow applications hosted in the cloud to send emails.

Amazon Simple Email Services (SES)

- It is a bulk and **transactional email-sending service from Amazon.**
- SES is an outbound- only email sending service that allows applications hosted in the Amazon cloud to send emails such as **marketing, transactional and other types of correspondence emails.**
- SES uses content filtering technologies to scan the outgoing email messages to help ensure that they do not contain material that is typically flagged as questionable by ISPs.
- SES used from the SES console, SMTP interface or SES API.

Application Services

Google Email Services

- It is part of the **Google App Engine** platform that allows App Engine applications to send email messages on behalf of the app's administrator and on behalf of users with Google Accounts.
- App engine Apps can also **receive emails**.
- Apps send messages using the **mail service** and **receive messages in the form of HTTP** requests initiated by App Engine and posted to the app.

4. Notification Services

- Cloud-based notification services or **push messaging service** allow applications to push messages to internet connected smart devices such as smartphones, tablets etc.
- It works based on publish- subscribe model in which consumers subscribe to various topics provided by a publisher.
- When ever new content is available on one of those topics , the notification service pushes that information out to the consumer.

Application Services

Amazon Simple Notification Services

- It is a **push messaging service** from Amazon.
- It has two types of clients like publishers and subscribers.
- Publishers communicate asynchronously with subscribers by producing and sending messages to topics.
- A topic is a logical access point or a communication channel.
- SNS can deliver notifications as SMS,email, or to SQS queues,or any HTTP endpoint.

Google Cloud Messaging GCM)

- GCM for android provides **push messaging** for android devices.
- GCM allows applications to **send data from the application servers** to their users, Androids devices and also to receive messages from devices on the same connection.
- GCM provides a send to sync message capability that can be used to inform an application to sync data from the server.

Application Services

Windows Azure Notification Hubs

- It is a **push notification service** from Microsoft that provides a common interface to send notifications to all major mobile platforms including windows store/windows phone8,Ios and Android.
- Platform specific infrastructures called ***Platform Notifications System(PNS)*** are used to deliver notification Messages.
- Devices register their PNS handles with the notification Hub.
- Each notification hub contains credentials for each supported PNS.
- These credentials are used to connect to the PNSs and send push notifications to the applications.

Application Services

5. Media Services

- Cloud service providers provide various media services for manipulating and transforming media such as images, videos and audio.

Amazon Elastic Transcoder

- It is a cloud based video transcoding service from Amazon.
- It is used to convert video files from their source format into various other formats that can be played on devices such as desktops, mobiles, tablets etc.
- It allows to perform multiple transcodes in parallel using transcoding pipelines.
- It works with the Amazon S3 for input and output video files are stored.

Application Services

Google Images Manipulation Services

- It is a part of the **Google App Engine** platform.
- Image Manipulation service provides the capability to resize, crop, rotate, flip and enhance images.
- It accepts the image data directly from the App Engine apps, from Google Blobstore or Google Cloud Storage.

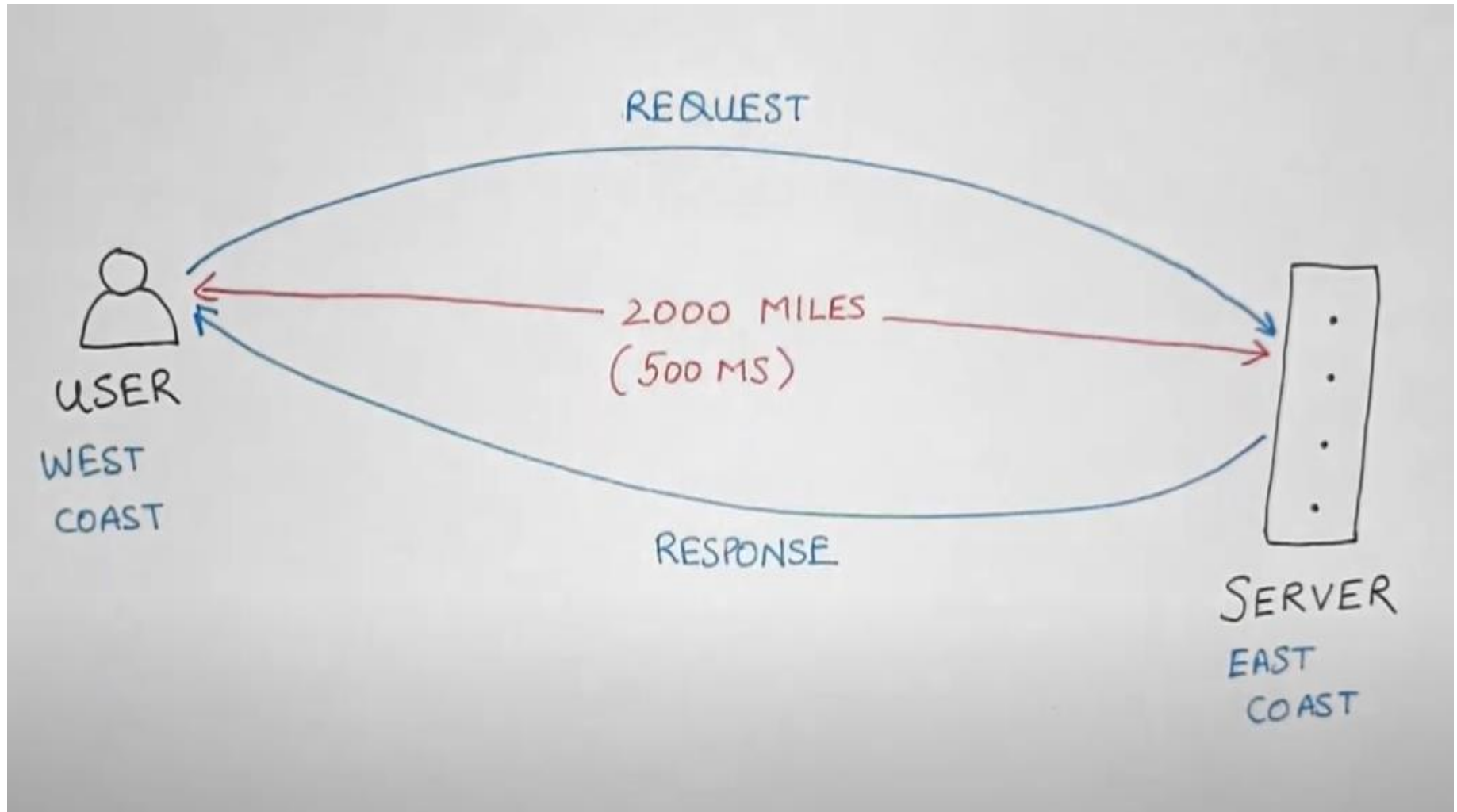
Windows Azure Media Services

- It provides the **various media services such as encoding and format conversion**, content protection and on-demand and live streaming capabilities.
- It provides applications the capability to build media workflows for uploading, storing, encoding, format conversion, content protection and media delivery.

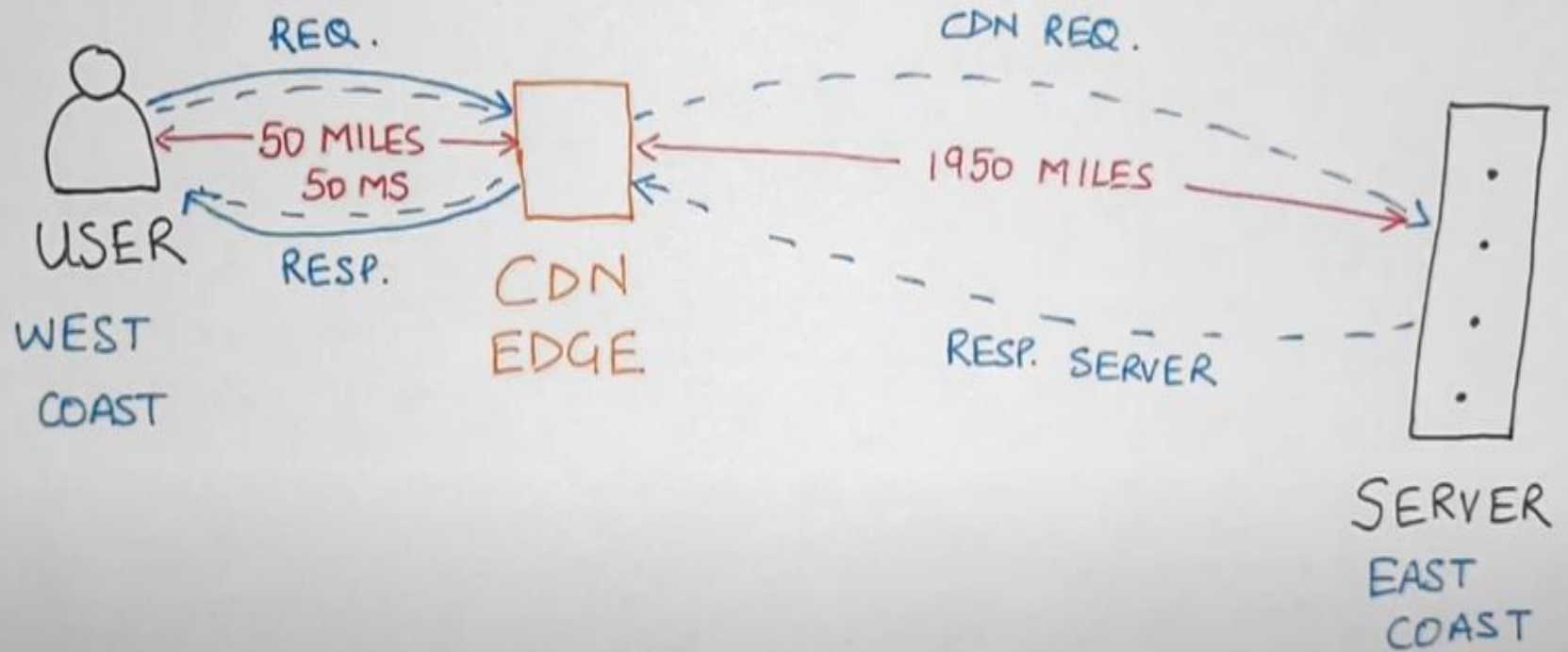
Content Delivery Services

- *Cloud based content delivery service* include content delivery networks(CDNs).
- A CDN is a *distributed system* of servers located across multiple *geographic locations* to serve content to end users with high availability and high performance.
- CDNs are useful for *serving static content like text, images, scripts* etc.. and streaming media.
- CDNs have a *number of edge locations deployed* in multiple locations, often over multiple backbones.
- Request for a static or streaming media content that is served by a CDN are directed to the nearest edge location.
- CDN catch the popular content on the edge servers which helps in reducing *bandwidth costs and improve response times*.

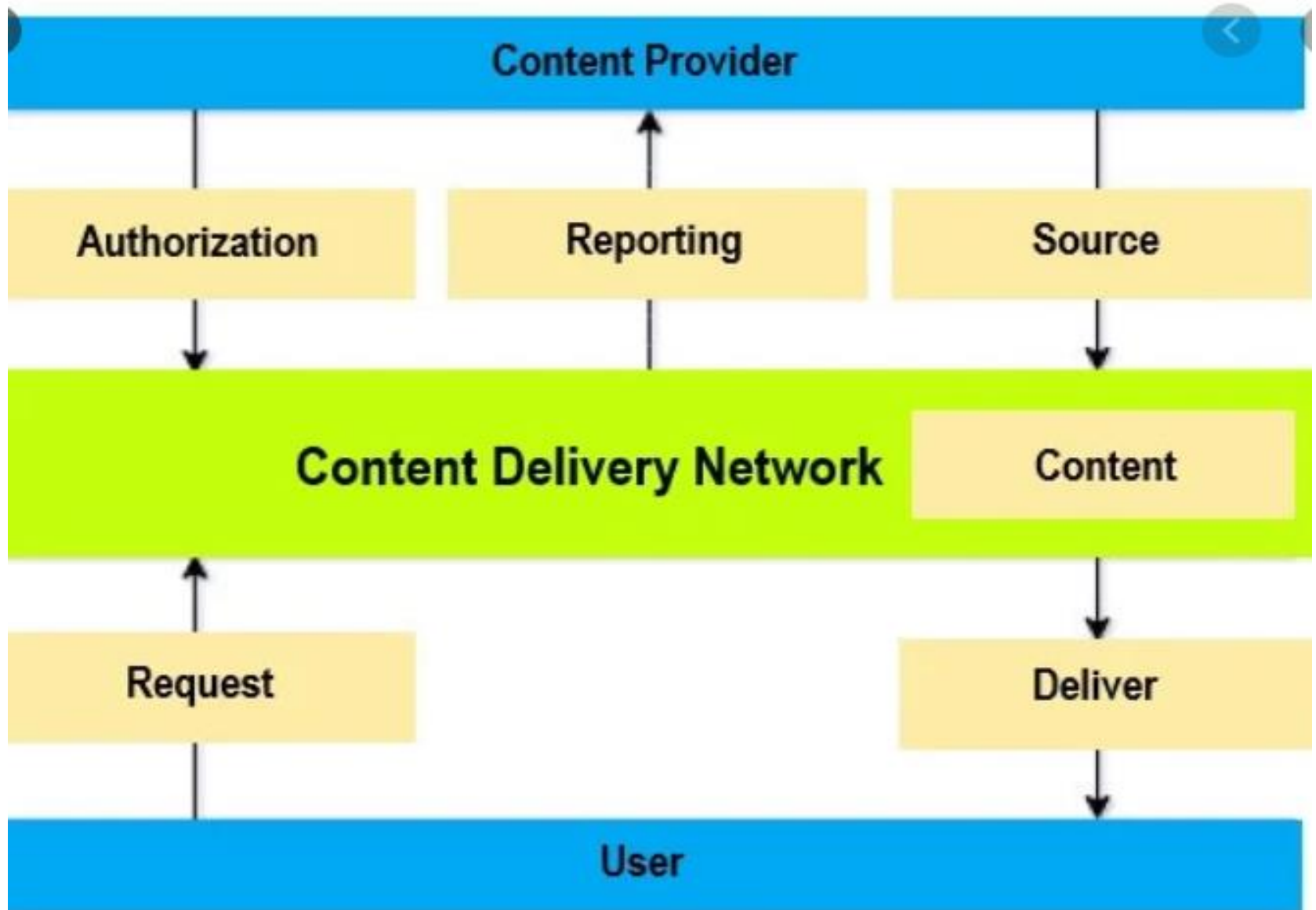
Content Delivery Services



Content Delivery Services



Content Delivery Services



- **Content:** digital information created and/or licensed for distribution

Content Delivery Services

- Content:** digital information created and/or licensed for distribution
- Content Provider:** the entity delivering the content
- Authorization:** the Content Provider gives the CDN Provider permission to deliver content
- Reporting:** the Content Provider demands performance analytics from the CDN Provider to evaluate the quality of the CDN Provider's service and have access to other relevant data
- Source:** the Content Provider sends a copy of the content
- Request:** the user requests from the Content Provider to view or locally store data (content)
- Deliver:** CDN delivers the content to the user
- User:** the entity requesting data (content) from the Content Provider

Content Delivery Services

Amazon Cloud Front

- It is a content **delivery** from Amazon.
- It is used to **deliver dynamic, static and streaming** content using a global network of edge locations.
- The content in the **cloudfront is organized** into distributions.
- Each distribution specifies the original location of the content to be delivered which can be an **Amazon S3 bucket, an Amazon EC2 instance, or an *Elastic Load balancer, or your own origin server.***
- Distributions can be accessed by their domain names.
- CloudFront helps in improving the performance of websites by:
 1. Catching the static content **like javascript, CSS and Images** etc. at edge locations.
 2. Proxing requests for **dynamic or interactive content back to the origin running** in the AWS cloud.

Content Delivery Services

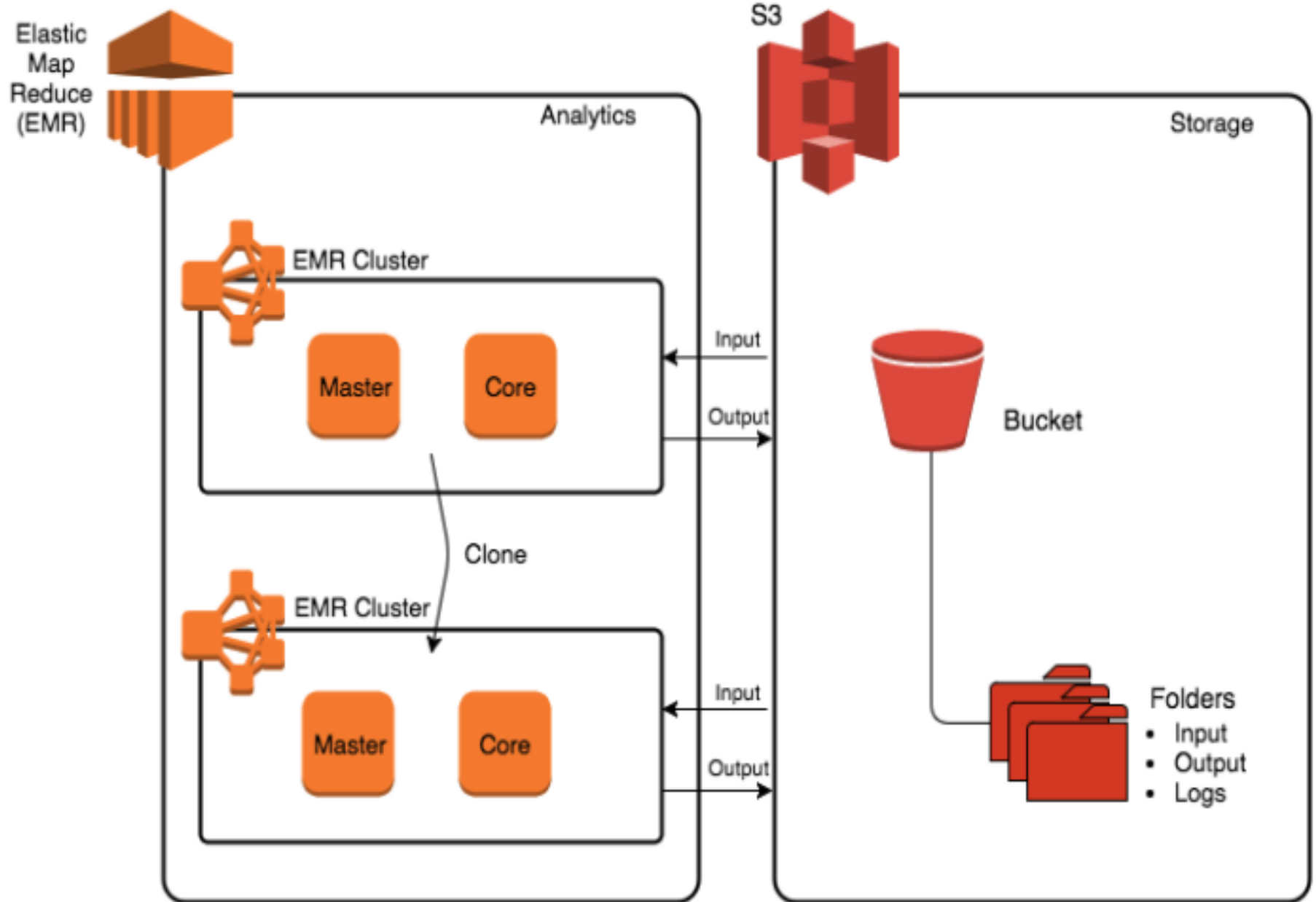
Windows Azure Content Delivery Network

- It is the Content Delivery Service from the Microsoft.
- Azure CDN caches windows Azure blobs (**Blob Storage Free. 25+ Free Services**) and static content at the edge locations (**Amazon CloudFront announces its first Edge locations in Kolkata. multiple Edge locations in Bengaluru, Chennai, Delhi, Hyderabad, and Mumbai.**) to improve the performance of the websites.
- Azure CDN can be Enabled on a windows Azure storage account.

Analytics Services

- It allow analyzing **massive data sets** stored in the cloud either in cloud storage or in cloud database using programming models such as **Map Reduce**.
- Using cloud analytics services applications can perform data-intensive tasks such as
 - Data mining
 - Log file analysis
 - Machine learning
 - Web indexing etc.

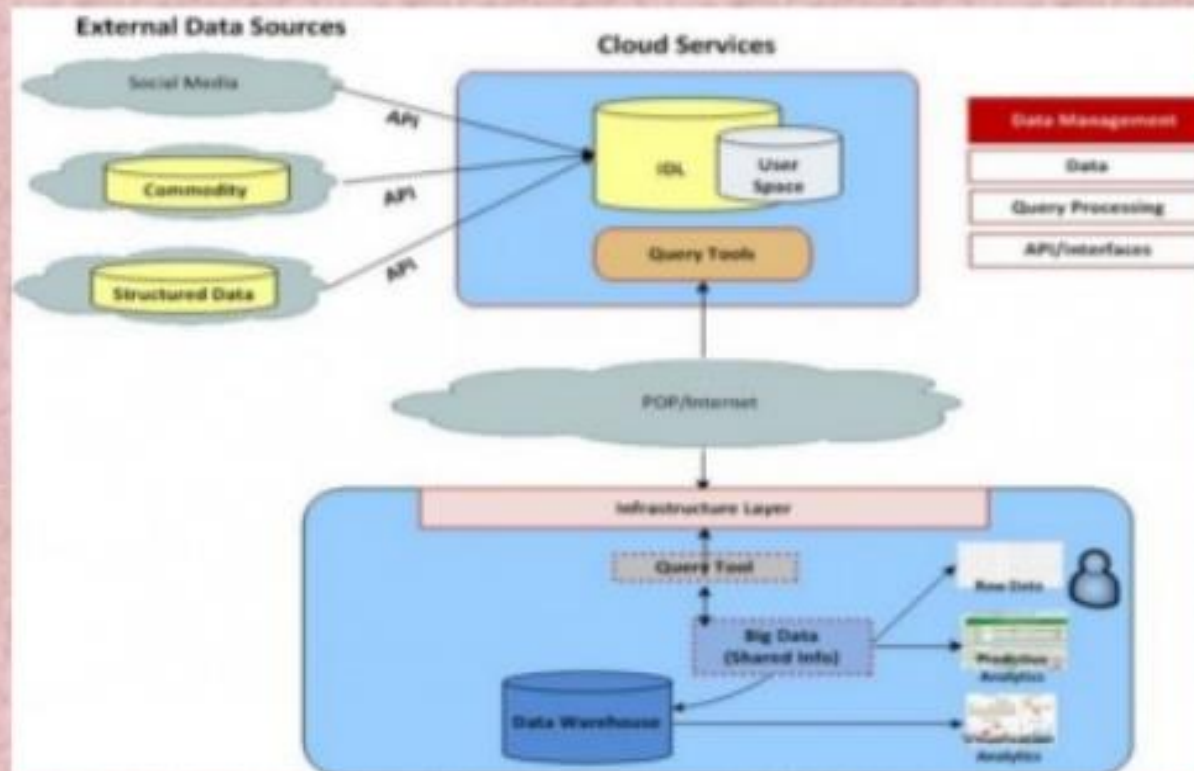
Analytics Services



Cloud Analytics

→ What is cloud analytics

→ Cloud analytics is a service model in which elements of the data analytics process are provided through a public or private cloud. Cloud analytics applications and services are typically offered under a subscription-based or utility (pay-per-use) pricing model.



→ Gartner defines the six key elements of analytics as data sources, data models, processing applications, computing power, analytic models and sharing or storage of results. In its view, any analytics initiative “in which one or more of these elements is implemented in the cloud” qualifies as cloud analytics.

Analytics Services

Amazon Elastic Map Reduce

- It is the Map Reduce service from Amazon based the Hadoop framework running on **Amazon EC2 and Amazon S3**.
- It supports different job types like :
 1. **Custom JAR :** **Custom JAR** job flow runs a java program that you have uploaded to Amazon S3.
 2. **Hive Program:** Hive is a data warehouse system for **Hadoop**. You can use Hive to process data using the SQL-like language, called **Hive-QL**. You can create a hive job flow with EMR which can be an interactive hive job or hive script.
 3. **Streaming Job:** Streaming job flow runs a single Hadoop job consisting of map and reduce functions implemented in a script or binary that you have uploaded to Amazon S3.

Analytics Services

You can write map and reduce scripts in ***ruby, perl, python, php, R, bash or C++.***

4. ***Pig programs:*** Apache pig is a platform for analyzing large data sets, that consists of a ***high-level language(Pig Latin)*** for expressing data analysis programs, coupled with infrastructure for evaluating these programs.
 - You can create a Pig job flow with EMR which can either be an interactive pig job or pig script.
5. ***Hbase:*** Hbases is a ***distributed, scalable, No-Sql*** database built on top of Hadoop.
 - EMR allows you to launch an ***Hbase cluster***. Hbase can be use for various purposes such as referencing data for Hadoop analytics, real-time long ingestion and batch log analytics.

Analytics Services

Google Map Reduce Services

- It is a part of the **map engine platform**.
- App Engine Map Reduce is optimized for App Engine Environment and **provides capabilities such as automatic sharding for faster execution**, standard data input readers for iterating over blob and data store data, Standard output writers etc.
- The MapReduce service can be accessed using the google MapReduce API.
- To Execute a **MapReduce** job a MapReduce pipeline object is instantiated within the app engine application.
- Map Reduce pipeline specifies the mapper, reducer, data input reader and output writer.

Analytics Services

Google BigQuery

- It is service for **querying massive datasets**.
- Big query allows querying datasets using **SQL-like queries**.
- The **BigQuery** queries are run against **append-only tables** and use the processing power of google's infrastructure for **speeding up queries**.
- To query data, it is first loaded into BigQuery using the BigQuery console or **BigQuery commandline tool or BigQuery API**.
- Data can be either in CSV or JSON format.
- The uploaded data can be queried using BigQuery's dialect.

Windows Azure HDInsight

- Windows Azure HDInsight is an analytic service from Microsoft.
- HDInsight deploys and provisions **Hadoop clusters** in the Azure cloud and makes Hadoop available as a service.
- HDInsight service uses Windows Azure Blob Storage as the default file system. It provides interactive consoles for both JS and Hive.

Deployment and Management Services

- It allow you to easily deploy and manage applications in the cloud.
- These Services automatically handle deployment tasks such as *capacity provisioning (equipment), load balancing, auto-scaling and application health monitoring*.

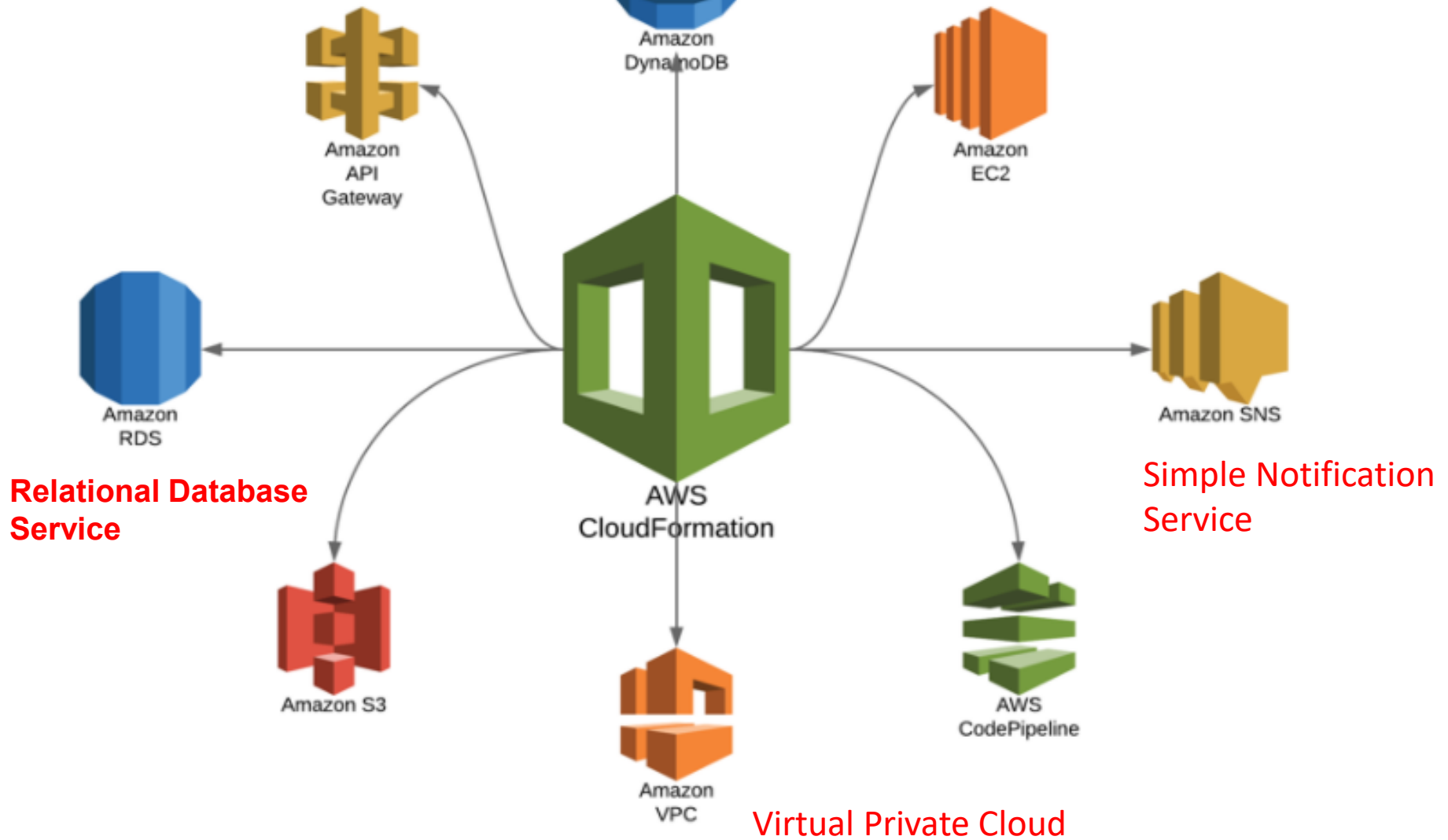
Amazon Elastic Beanstalk

- Amazon provides a deployment service called Elastic Beanstalk that allows you to *quickly deploy and manage applications in the AWS cloud*.
- Elastic Beanstalk supports *Java, PHP, .NET, Node.js, Python and Ruby applications*.
- In Elastic Bean Stalk you just need to upload the application and specify configuration settings in a simple wizard and the service automatically handles instance *provisioning, server configuration, load balancing and monitoring*.

Deployment and Management Services

Amazon Cloud Formation

- Amazon Cloud Formation is a **deployment** management service from Amazon.
- With cloud Front you can create deployments from a collection of AWS resources such as **Amazon Elastic Compute Cloud, Amazon Elastic Block Store, Amazon Simple Notification Service, Elastic Load Balancing and Auto Scaling.**
- A collection of AWS resources that you want to manage together are organized into a **stack**.
- Cloud Formation Stacks are created from cloud formation templates.
- It allows to create your own templates or use the predefined templates.



Amazon Cloud Formation

Identity and Access Management Services

- *IDAM* services allow managing the *authentication and authorization of users to provide secure access to cloud resources*.
- These are useful for organizations which have *multiple users* who access the cloud resources.
- By using IDAM services you can manage user identifiers, user *permissions, security credentials and access keys*.

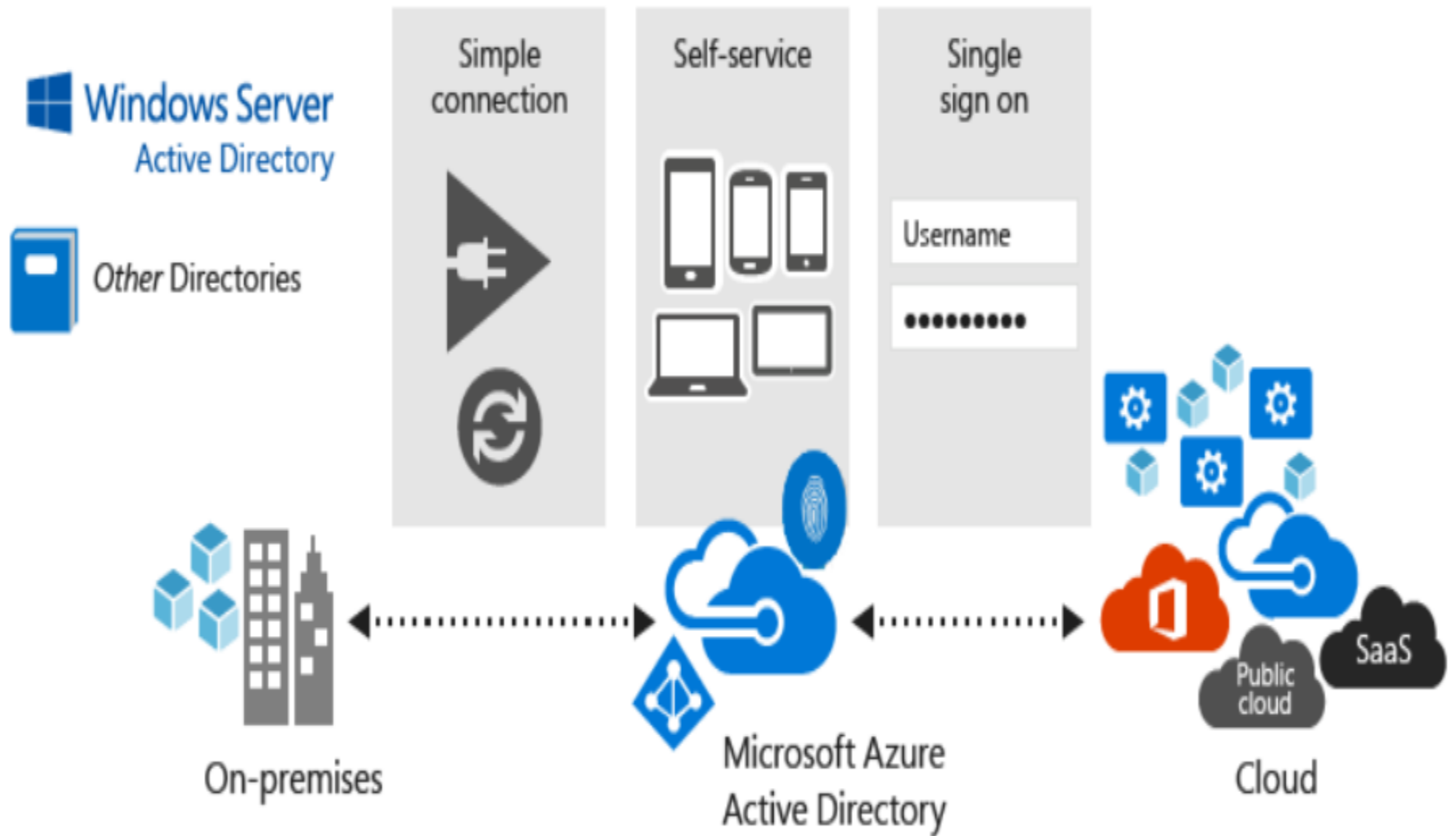
Amazon Identity and Access Management

- AWS IDAM allows you to manage users and user *permissions for an AWS account*.
- With IDAM you can manage *users, security credentials such as access key and permissions that control which AWS resources users can access*.
- Using IDAM you can control what data users can access and what resources users can create.
- It allows creation, rotation and revocation of user credentials.

Identity and Access Management Services

Windows Azure Active Directory

- It is an IDAM service from Microsoft.
- It provides a **cloud based identity provider** that easily integrates with your on-premises active directory deployments and also provides **support for third party identity providers**.
- With Azure **Active directory** you can control access to your **applications in Windows Azure**.

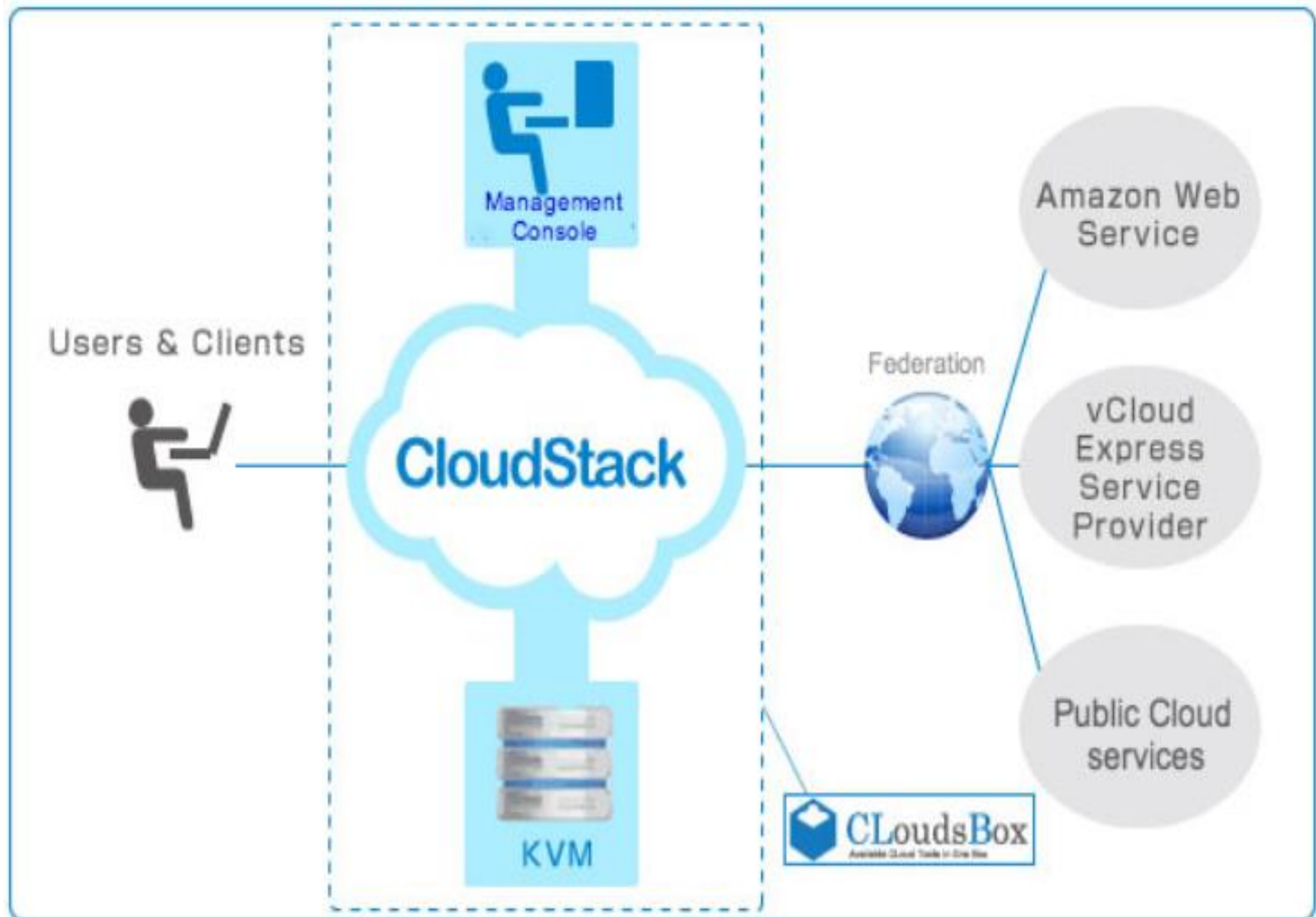


Windows Azure Active Directory

Open Source Private Cloud Software

1. Cloud Stack

- **Apache CloudStack** is an opensource cloud software that can be used for **creating private cloud offerings**.
- **CloudStack** manages the **network, storage and compute nodes that make up a cloud infrastructure**.
- A Cloud Stack installation consists of a Management Server and the cloud infrastructure that it manages.
- The cloud infrastructure can be as simple as one host running the hypervisor or a **large cluster of hundreds of hosts**.
- The management server allows you to **configure and manage the cloud resources**.
- The Management server **manages one or more zones** where each zone is typically a single datacenter.
- Each zone has one or more pods. **A pod is a rack of hardware comprising of a switch and one more clusters**.



Open Source Private Cloud Software

- A cluster consists of one or more hosts and a primary storage.
- A host is a compute node that runs guest virtual machines.
- The Primary storage of a cluster stores the disk volumes for all the virtual machines running on the hosts in that cluster.
- Each zone has a secondary storage that stores templates, ISO images and disk volume snapshots.
- A region is a specific geographical location where you can run your resources.

Open Source Private Cloud Software

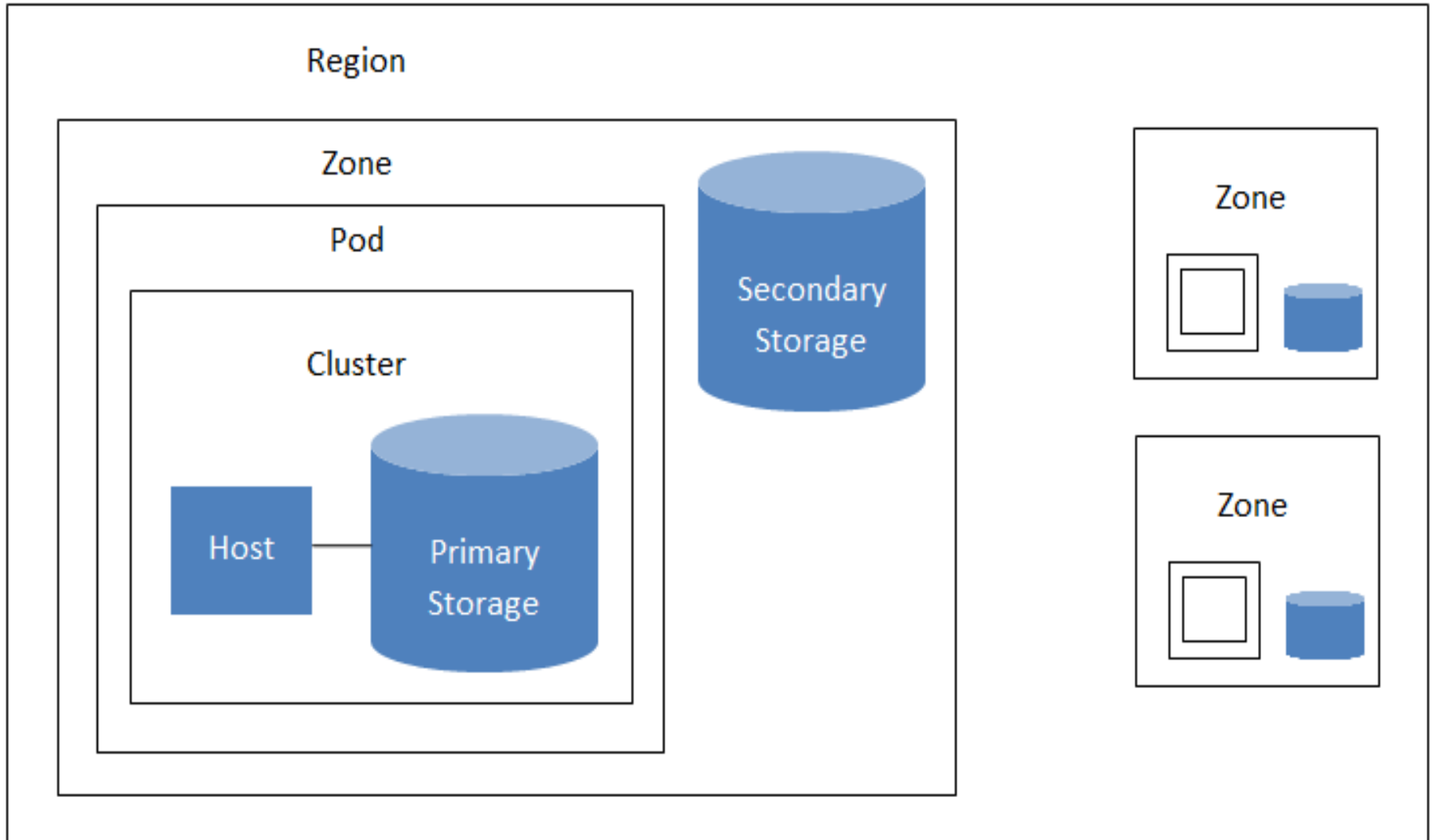


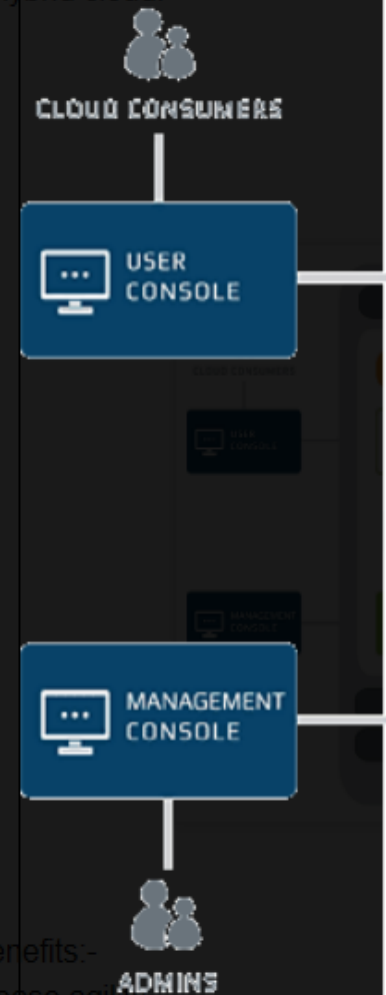
Fig : Cloud Stack Architecture

Open Source Private Cloud Software

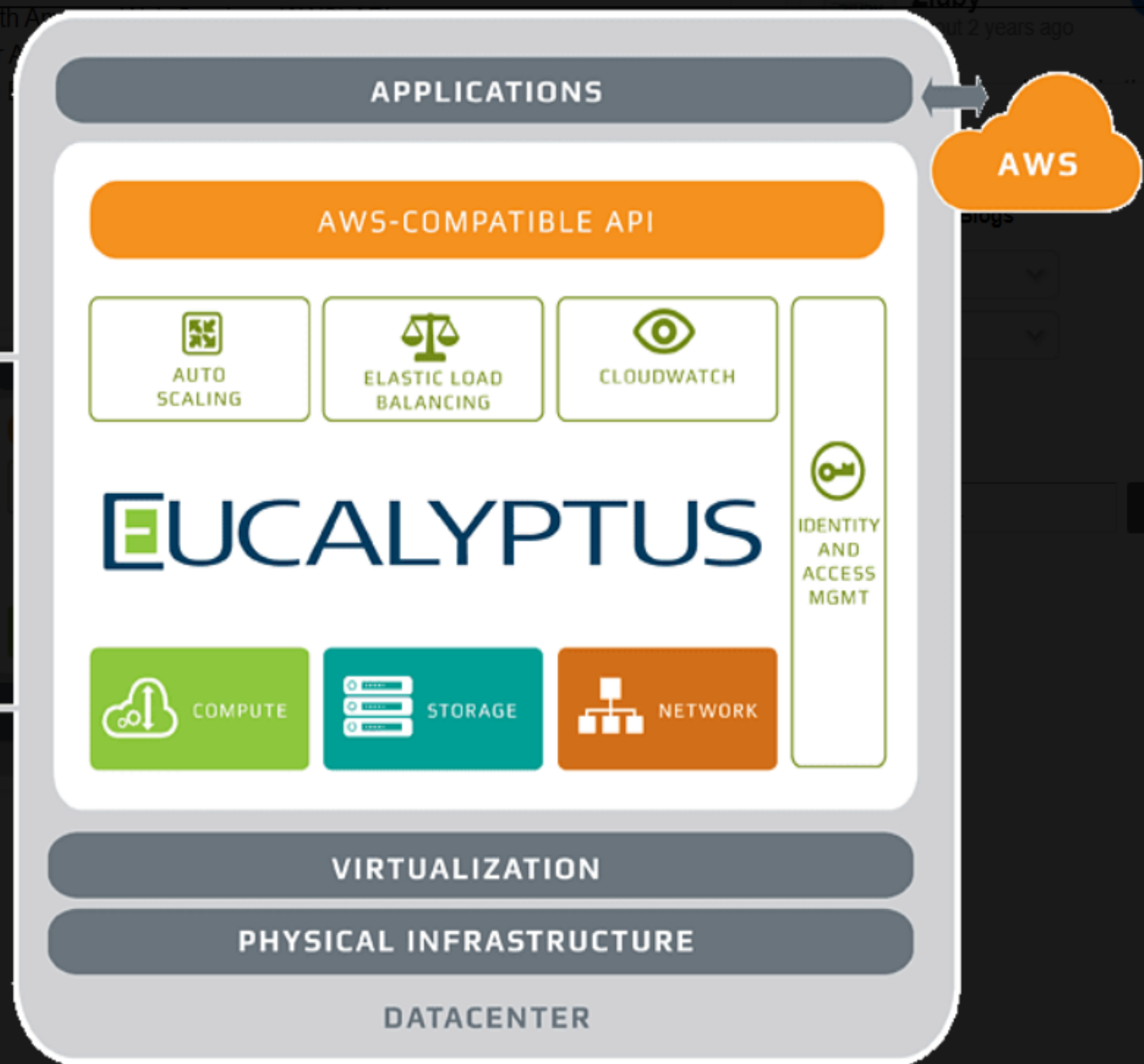
2. Eucalyptus

- Eucalyptus is an *open source private cloud software for building private and hybrid clouds* that are compatible with Amazon Web Services(AWS) APIs.
- The **Node controller** hosts the virtual machine instances and manages the virtual network endpoints.
- The cluster level consists of three components like *cluster controller(CC), Storage Controller(SC) and VMWare Broker*.
- The CC manages the **Virtual machines** and is the front end for a cluster.
- The SC manages the Eucalyptus block volumes and snapshots to the instances within its specific cluster.
- SC is similar to AWS Elastic Block Store(EBS).
- VMWare broker is an optional component that provides an AWS – compatible interface for VMWare environments.

est part is, it is compatible with Amazon
we can manage Eucalyptus or Amazon
move their instances between
e hybrid cloud.



Benefits:-
Increase agility
Lower Cost
Improve Security



Open Source Private Cloud Software

- At the cloud level there are **two components** like Cloud Controller (CLC) and Walrus.
- CLC provides an **administrative interface** for management and performs high level resource scheduling, **system accounting, authentication and quota management.**
- Walrus is equal to **Amazon S3** and serves as a **persistent** storage to all of the virtual machines in the Eucalyptus cloud.

Open Source Private Cloud Software

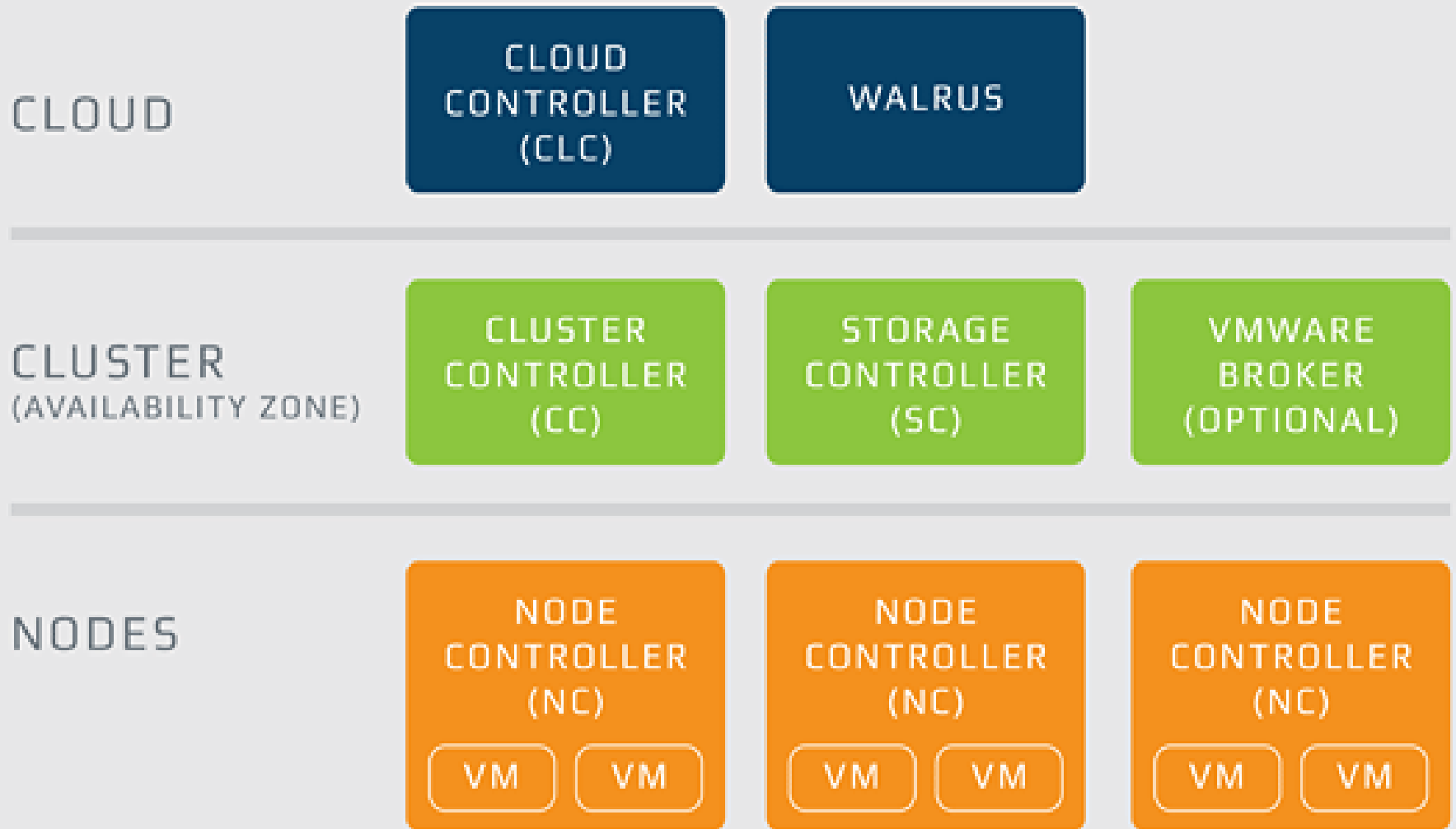
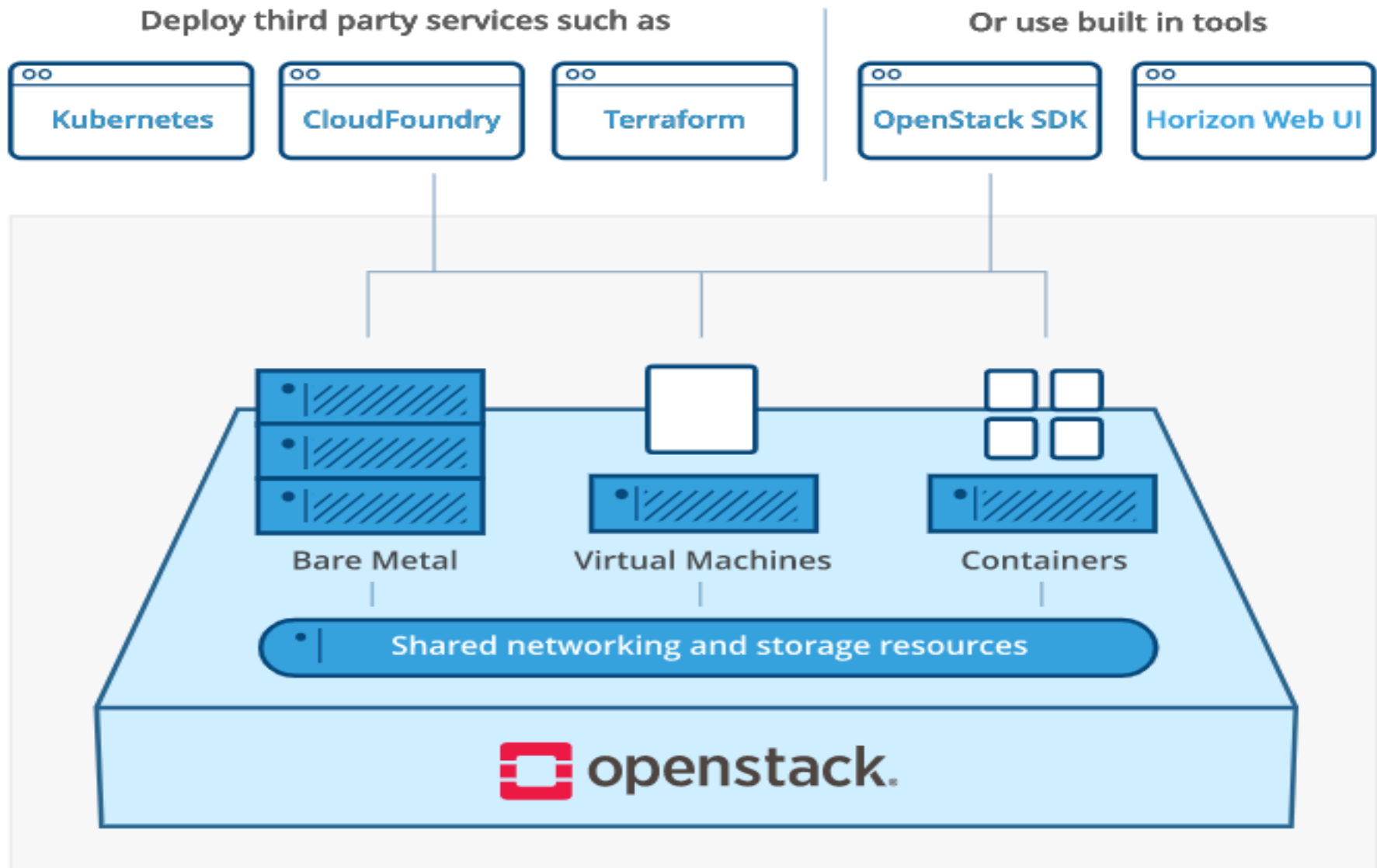


Fig: Eucalyptus Architecture

Open Source Private Cloud Software

3. OpenStack

- OpenStack is a **cloud operating system** comprising of interactive services that control computing, storage and networking resources.
- The OpenStack compute service (**nova-compute**) manages networks of virtual machines running on nodes, providing virtual servers on demand.
- The network service(**nova-networking**) provides connectivity between the interfaces of other OpenStack services.
- The volume service (**Cinder**) manages storage volumes for virtual machines.
- The object storage service (**swift**) allows users to store and retrieve files.
- The identity service (**Keystone**) provides authentication and authorization for other services.
- The image registry(**glance**) acts as a catalog and repository for virtual machine images.



OpenStack is a cloud operating system that controls large pools of compute, storage, and networking resources throughout a datacenter, all managed and provisioned through APIs with common authentication mechanisms.

Open Source Private Cloud Software

- The open stack scheduler (**nova –scheduler**) maps the nova-API calls to the appropriate OpenStack components.
- The scheduler takes the virtual machine requests from the queue and determines where they should run.
- The messaging service(**rabbit-mq**) acts as a central node for message passing between daemons.
- Orchestration activities such as running an instance are performed by the nova-api which accepts and responds to end user compute API calls.
- The OpenStack dashboard(**called horizon**) provides web-based interface for managing OpenStack service.

Open Source Private Cloud Software

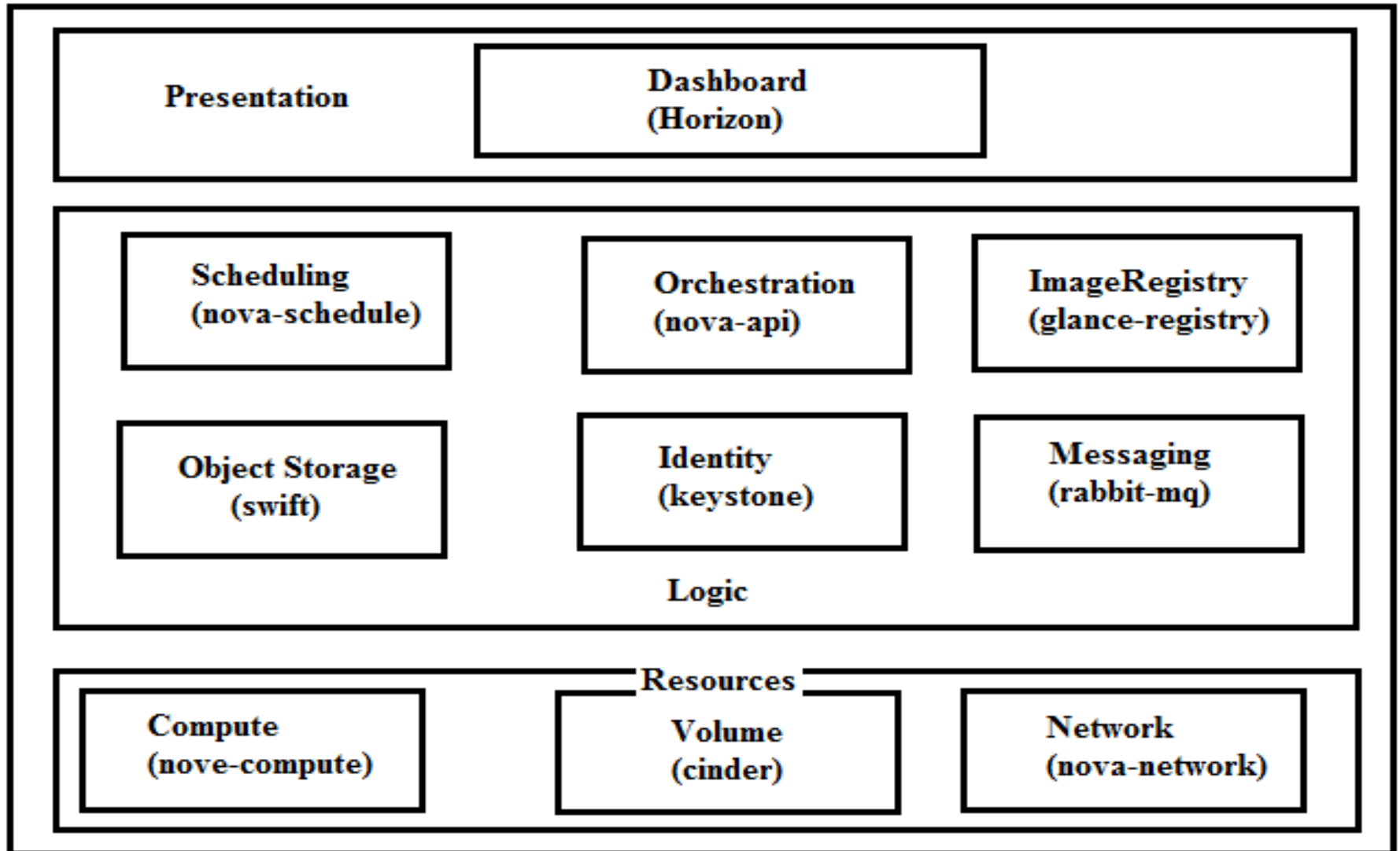


Fig : OpenStack Architecture

Thank You